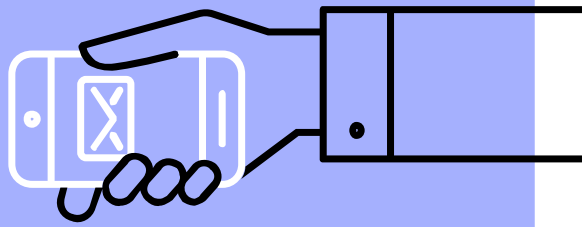
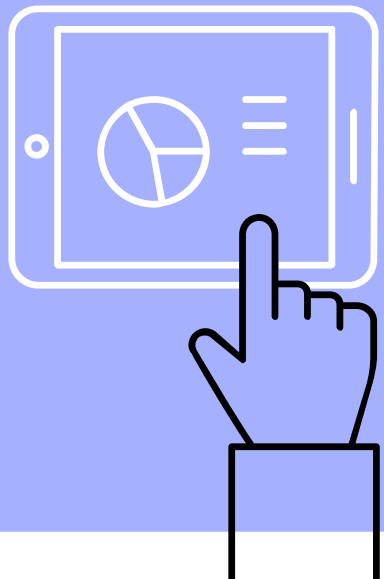
Growth Modeling & Data-Driven Strategy



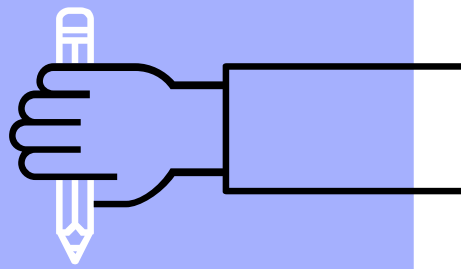




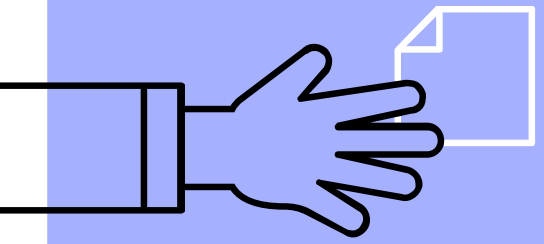
HELLO!

I am Aurore Kervella

15 years building growth systems at Dior, Sephora, Uber, Heetch and Meetic - from data to strategy.



1.
THE ARCHITECTURE OF GROWTH



Funnels describe
Loops drive

Funnels

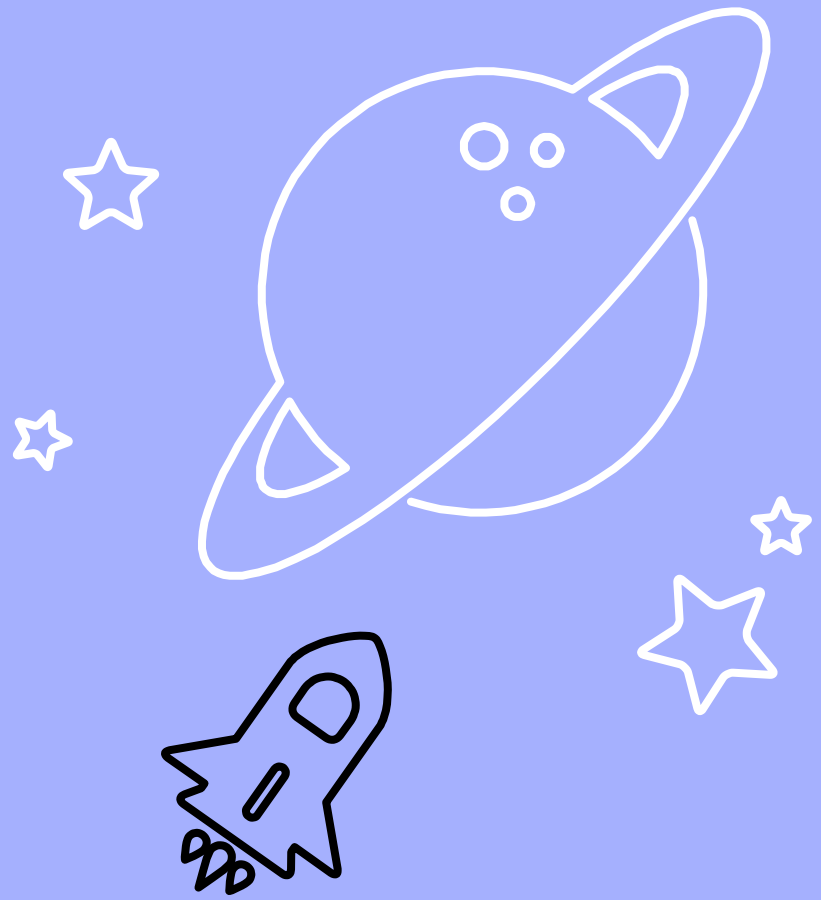
the anatomy of conversion

Loops

the mechanics of compounding

North Star Metric

the guiding light



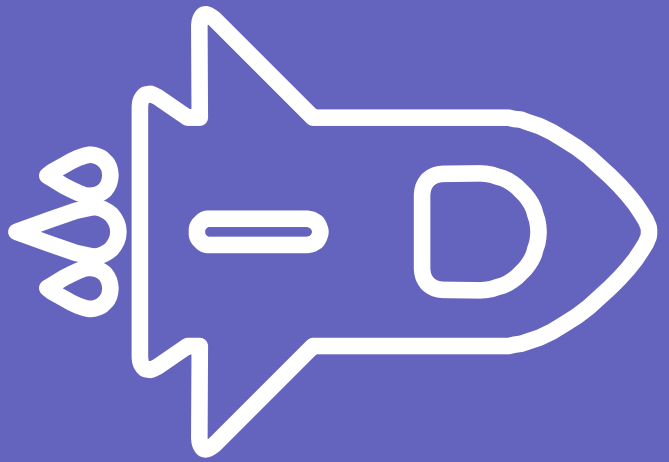
WHAT IS GROWTH REALLY?

Marketing

Loops

Data

Product





Growth

=

increase in users, revenue, or engagement over time



WHAT IS GROWTH?

- **Linear** = fixed increments

  Steady, predictable increases in fixed increments. Think traditional advertising - spend more, get proportionally more results.

  $y = a + b * x$

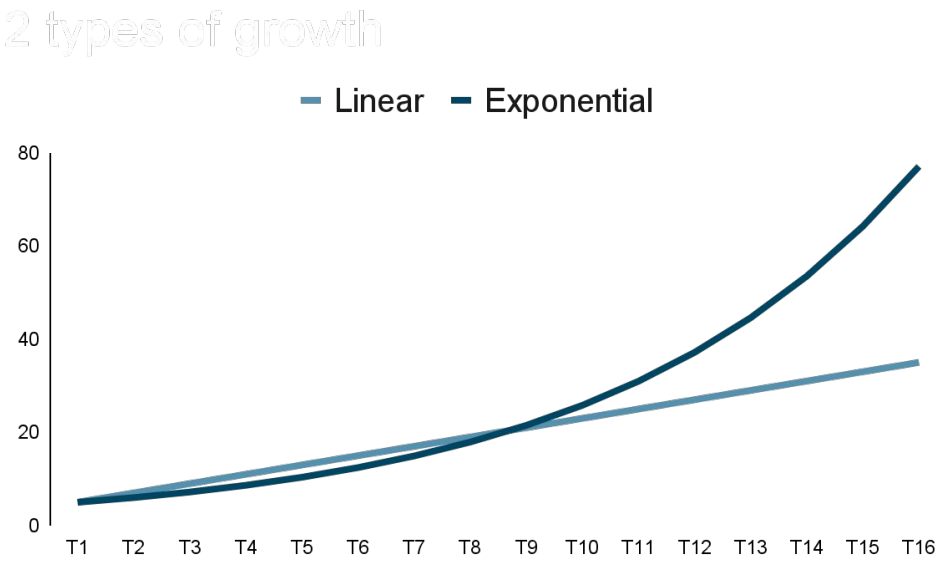
- **Exponential** = compounding

  Compounding effects where growth accelerates over time. Each new user potentially brings more users.

  $y = a * b \wedge x$



2 TYPES OF GROWTH



"Why do most startups seek exponential rather than linear growth? What makes it harder to achieve?"

THE ENGINE OF GROWTH

- **Funnels**
  A multi-step process (e.g. AARRR) where users drop off at each step. Requires **continuous input** (marketing spend) at the top.

- **Loops**
  A closed system where the **output** of one cycle (e.g., a retained user) directly feeds the **input** of the next cycle (e.g., new user acquisition). Drives compounding growth.



UNDERSTANDING FUNNELS

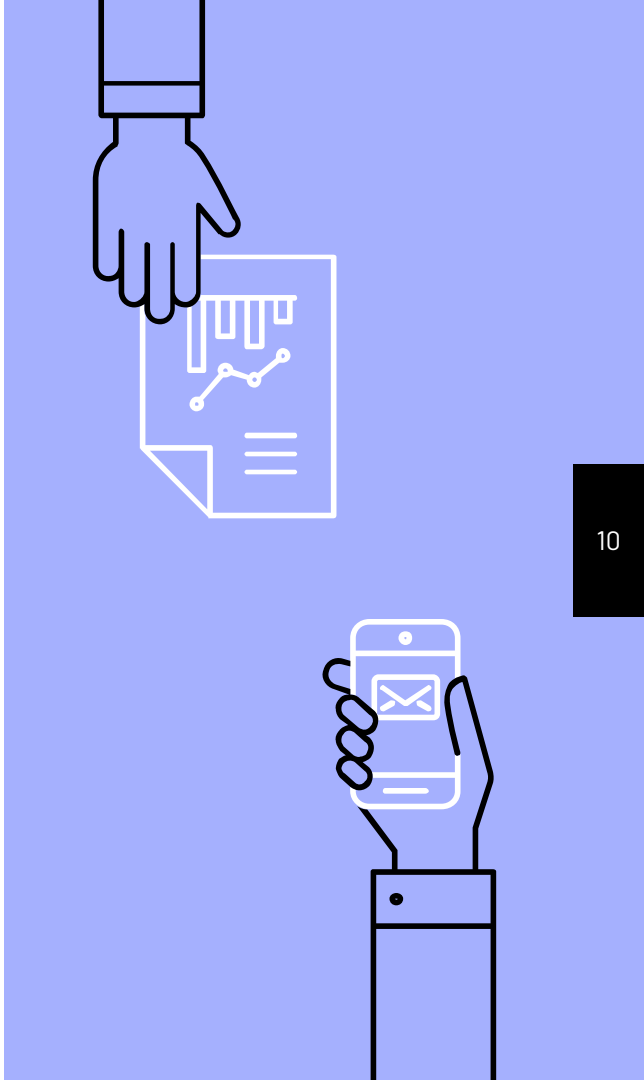
The traditional linear approach to converting prospects into customers through sequential stages.

Awareness
Potential customers discover your product

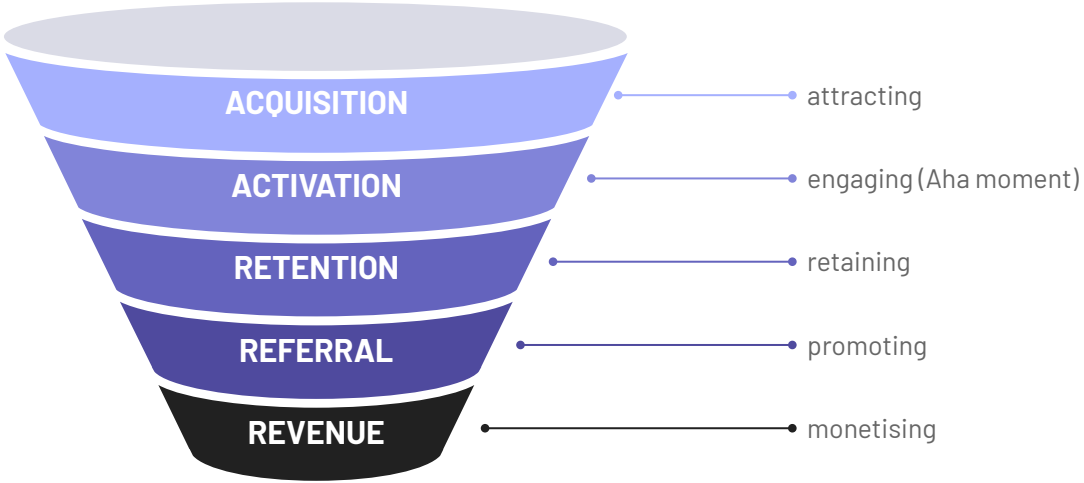
Consideration
Users evaluate your solution against alternatives

Conversion
Prospects become paying customers

Key Focus: Optimizing efficiency and identifying where users drop off in the process.

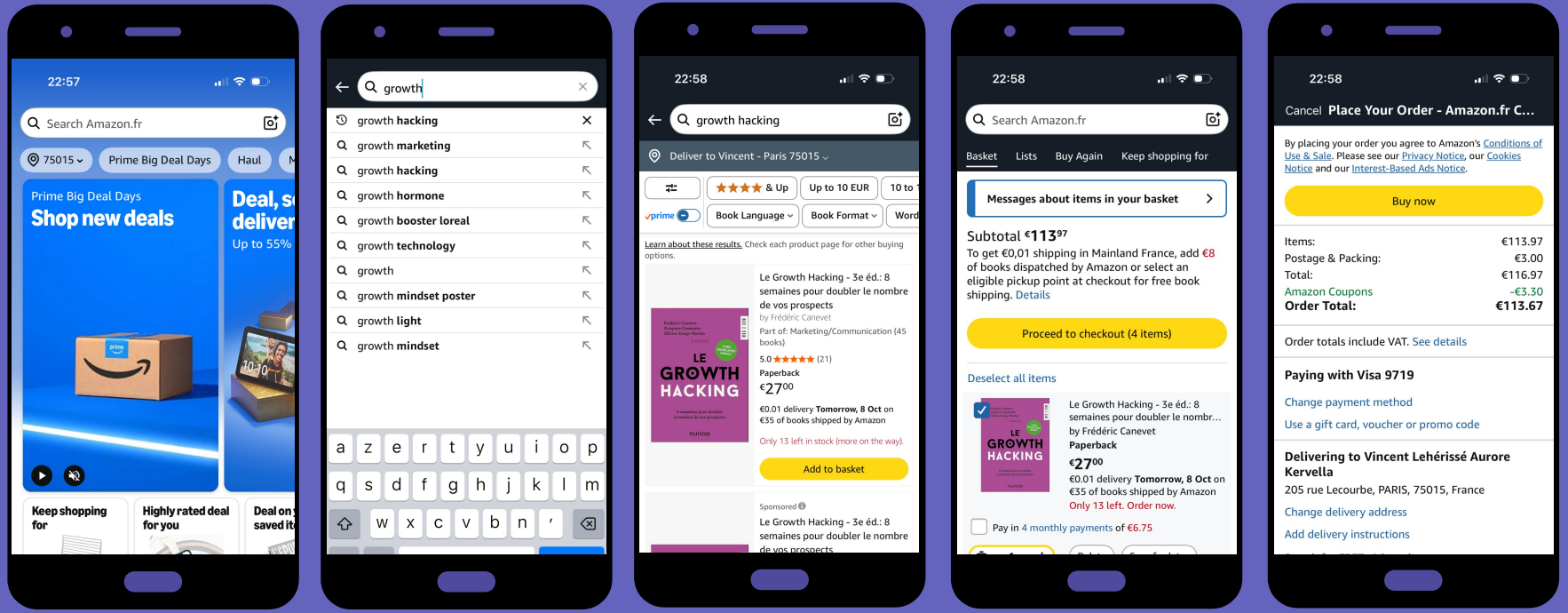


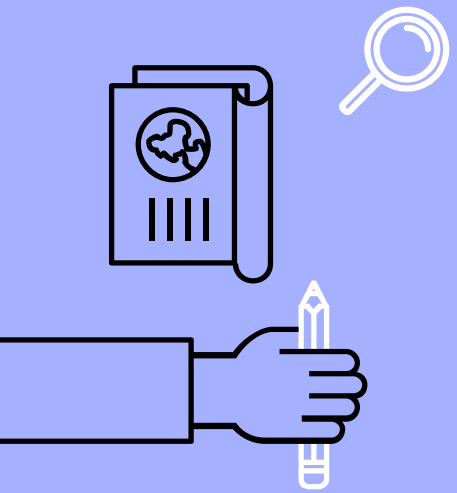
THE CLASSIC FUNNEL : AARRR



Conversion × Retention × Reactivation = Growth

Funnel example : Amazon





Funnel exercise

- You’re launching a new product : a learning app (Duolingo like)
- Your objective is to **convert users into paying subscribers**
- Define the **4 key steps** of the growth funnel from acquisition to retention / revenue



Funnels measure efficiency

- Linear
- Descriptive
- Optimizes conversion
- Ends with sale



Loops create momentum

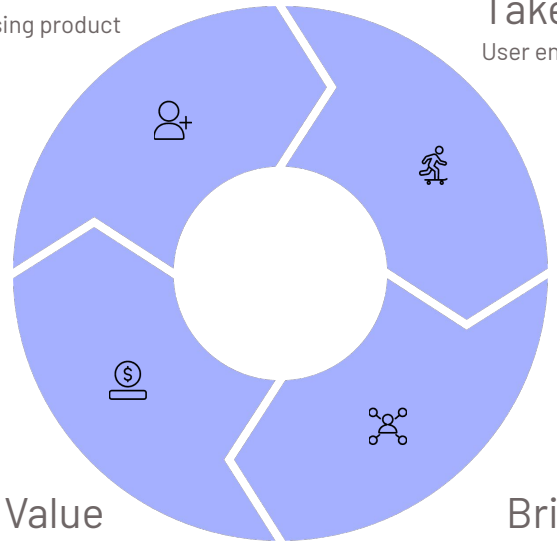
- Circular
- Generative
- Sustains momentum
- Feeds itself

GROWTH LOOPS EXPLAINED

Self-reinforcing systems where existing users drive acquisition of new users, creating compounding growth.

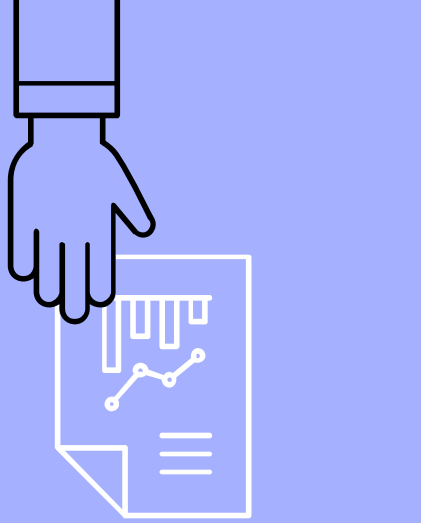
User Joins
New customer starts using product

Takes Action
User engages with core features



Creates Value
More users = more value for everyone

Brings New Users
Action naturally involves others





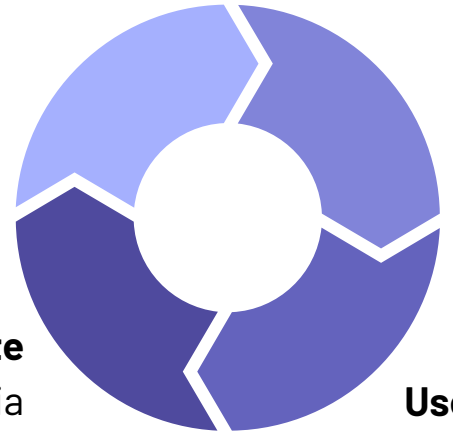
Type	Core Mechanism	Example	Key Metric(s)	Underlying Psychology / Engine
Acquisition Loop	Every user helps bring in *new users* — direct virality or content exposure.	**Dropbox** → Users invite friends to earn free storage. **TikTok** → Every piece of content shared externally drives new installs.	**K-Factor**, Invite Rate, Conversion Rate	Social proof, reciprocity, status — people share because it gives them utility or identity.
Retention Loop	Users stay engaged because *each action increases their own future value* — the more they use, the harder it is to leave.	**Duolingo** → Streaks, leaderboards, progress tracking. **Slack** → The more messages your team sends, the more integrated it becomes.	**DAU/MAU**, Churn Rate, Cohort Retention	Habit formation, sunk cost effect, community belonging.
Monetization Loop	User activity or purchases make the product *more valuable for others or more likely to generate revenue again.*	**Amazon** → Reviews → Trust → More Sales → More Reviews. **Freemium SaaS** → Free users convert → Revenue funds better product → More users convert.	**ARPU**, Conversion Rate, Net Revenue Retention (NRR)	Trust, network effect, economic flywheel.



Loop example : Dropbox

Dropbox referral program (extra storage for both inviter and invitee)

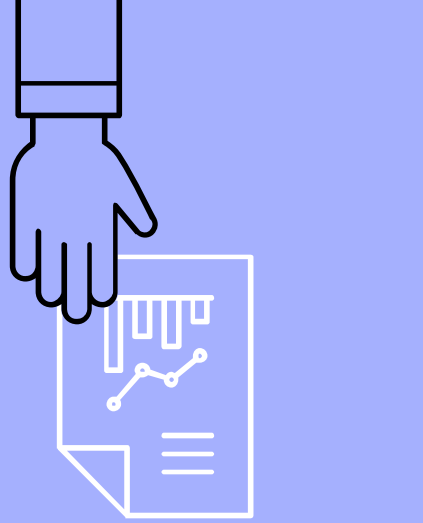
User signs up
start storing and sharing their files



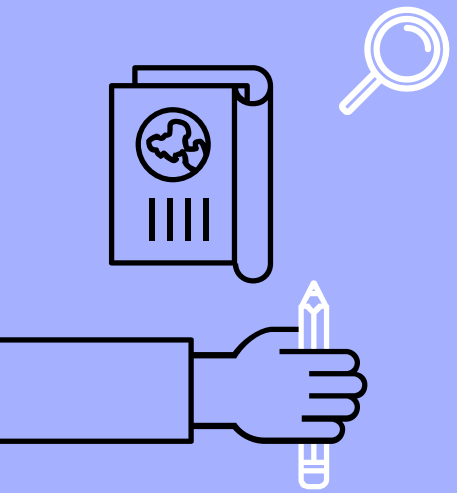
Incentive to share
Extra storage if they refer their friends to join

New user invite
referral invites sent via email or social media (nice and easy)

User select contacts
Invites are sent

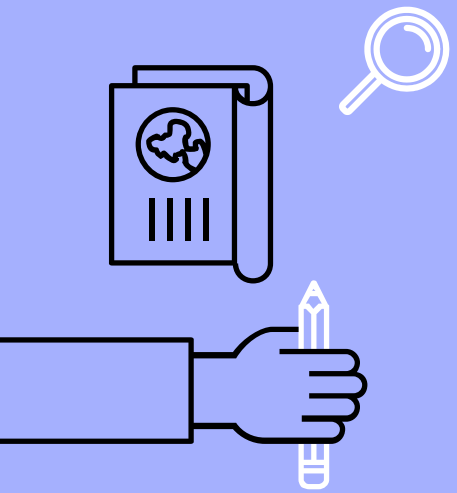






Loop exercise

- Design the primary growth loop for a platform
- We will use **LinkedIn** as our case study
  - Identify the Value
  - Identify the Input / Output
  - Map the Steps on the loop



Loop vs Funnel exercise

- Identify one funnel-driven company and one loop-driven company
- Share with the group



North Star Metric

(NSM)

=

A single metric that best captures the core value delivered to users.



THE NORTH STAR METRIC

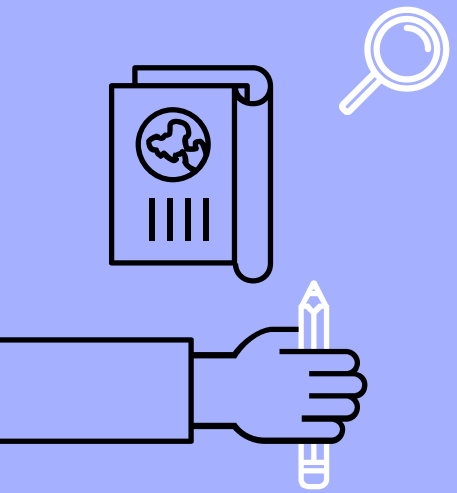
- **Aligns** cross-functional teams.
- Focuses on **sustainable, long-term growth** instead of vanity metrics.
- Typically a retention or engagement metric of some kind.
  - Daily active users (collaboration apps)
  - Monthly active subscribers (subscription)
  - Marketplace businesses : chicken or egg problem : usually supply side



THE NORTH STAR METRIC

- Examples :
  - Airbnb → Nights booked.
  - Spotify → Time spent listening.
  - Slack → Messages sent per team.

WhatsApp = Messages Sent = $\frac{\text{Total Messages Sent}}{\text{Daily Active Users}}$ = $\frac{\text{(New messages sent) + (Message Replies)}}{\text{(New Users) + (Returning Users)}}$

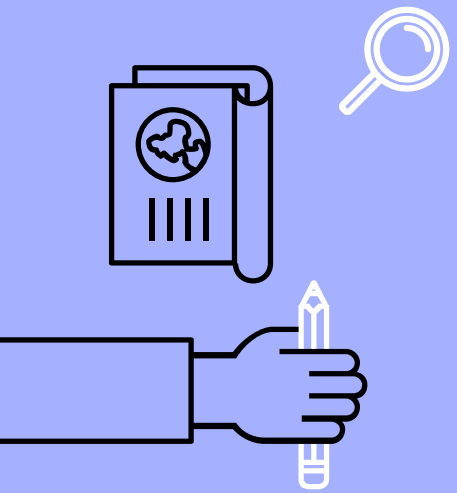


NSM exercise

- Revisit your team’s funnel or loop.
- What metric best reflects real value creation?
- Why would aligning everyone around it help?
- Share with the group

MINI RECAP BOARD

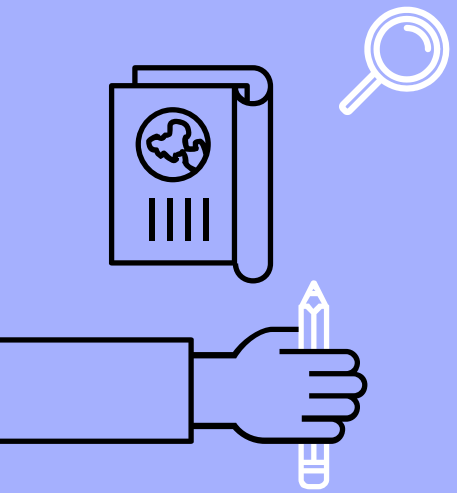
Concept	Purpose	Risk
Funnel	Diagnose journey	Over optimize micro steps
Loop	Driver compounding	Hard to identify early
NSM	Align teams	Tunnel vision



Time for a quiz (1/3)

You’re analyzing a startup that spends heavily on ads to drive installs.
Once users sign up, very few invite others, and growth only happens when ad spend increases.
Which growth architecture does this company use?

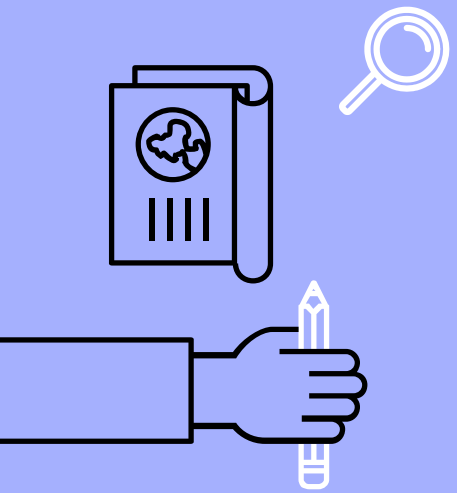
A. Loop
B. Funnel
C. Viral engine
D. Product-led system



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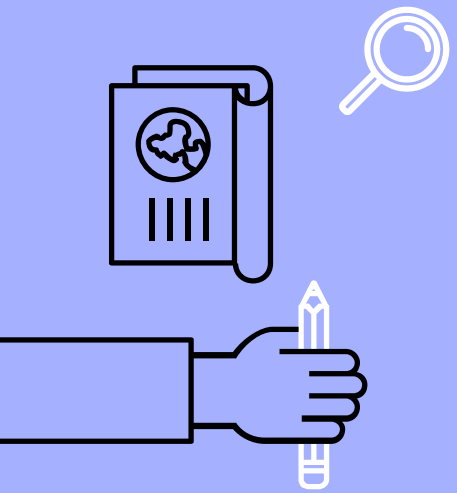
A. Loop
B. Funnel
C. Viral engine
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Time for a quiz (2/3)

Which of the following is the *best* North Star Metric for a food-delivery platform?

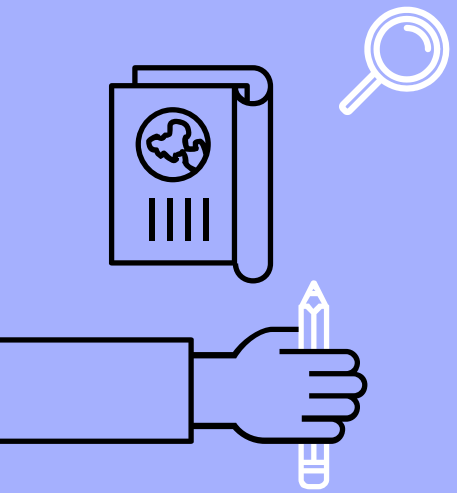
A. Number of app downloads
B. Total marketing spend
C. Completed deliveries per active customer
D. Number of push notifications sent



Time for a quiz (2/3)

Which of the following is the *best* North Star Metric for a food-delivery platform?

A. Number of app downloads
B. Total marketing spend
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D. Number of push notifications sent



Time for a quiz (3/3)

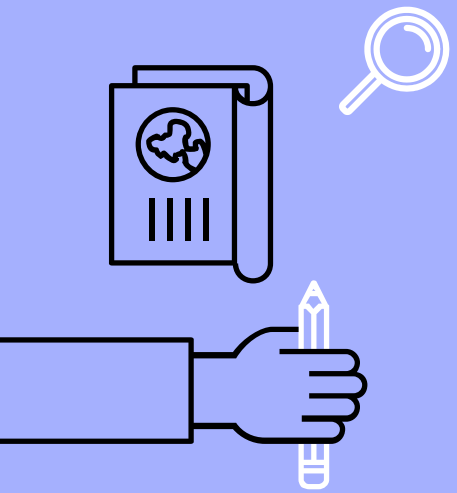
Dropbox famously grew through a referral program where users got extra storage for inviting friends. Which element made this a true growth loop rather than just a marketing tactic?

A. Each new user could trigger the same invite behavior.

B. Invited friends were automatically subscribed.

C. The reward was financial.

D. Dropbox ran TV ads about it.



Time for a quiz (3/3)

Dropbox famously grew through a referral program where users got extra storage for inviting friends. Which element made this a true growth loop rather than just a marketing tactic?

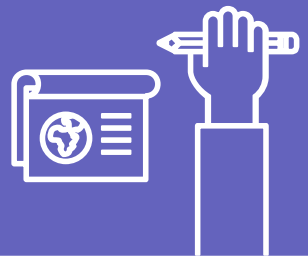
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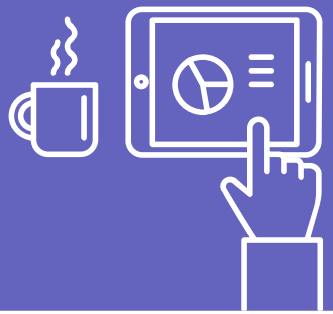
Which part of a growth system is hardest to measure?

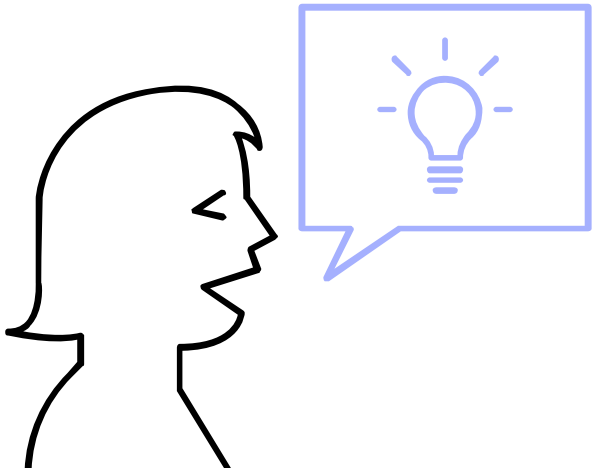




Can a company survive without loops?

When does a North Star conflict with user trust?





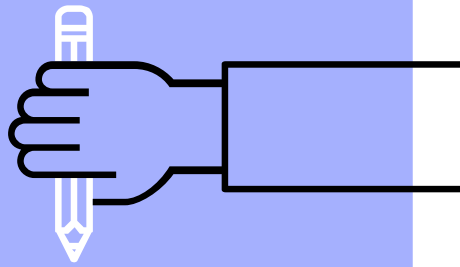
Growth can be linear or exponential

Loops drive compounding

Funnels = efficiency

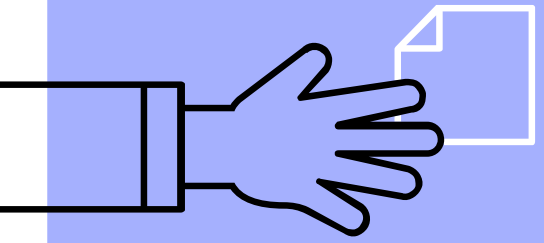
Loops = sustainability

NSM aligns teams and ensures value creation



2.
METRICS THAT MATTER

From data to decision



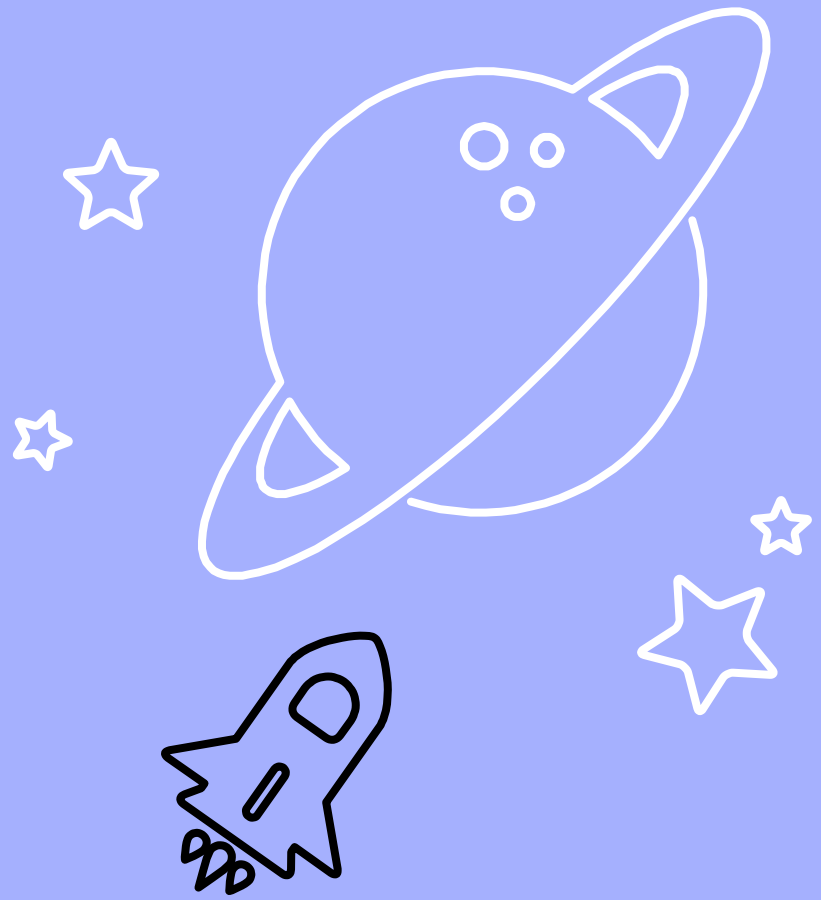
CAC & LTV

cost and revenue

Churn & Retention

How to retain a user

Virality & Word of Mouth



WHO IS REALLY GROWING?

Startup A



grows fast with paid ads (€1 M burn)

Startup B



grows slowly but retains 80 % monthly



Growth Metrics

(CAC, LTV,...)

=

Without the right metrics, growth looks like magic. With them, it becomes a science



CAC (cost of acquisition)

How much does it cost on average to acquire a new customer?

$$CAC = \frac{Total\,Acquisition\,Spend}{New\,Customers}$$

- Example :
  - Total spend: $10,000
  - New customers: 500
  - **CAC** = $10,000 / 500 = **$20 per customer**



LTV (lifetime value)

The total net revenue you expect to earn from a customer before they churn.

$$LTV = \text{ARPU} \times \text{Gross Margin} \times \text{Customer Lifetime}$$

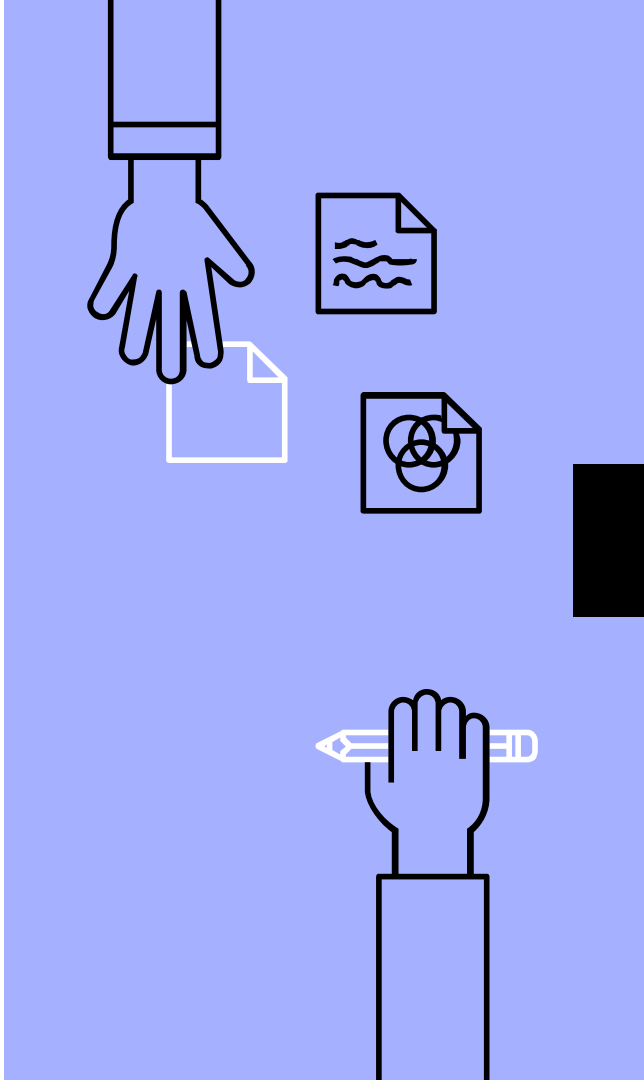
- Definitions :
  - **ARPU** = Average Revenue per User
  - **Gross Margin** = % of revenue retained after costs
  - **Customer Lifetime** ≈ 1 / monthly churn
  - **In quizzes** (margin = 100 %): LTV = ARPU / churn



EXAMPLE

- Total spend: $20,000
- New customers: 500
- **CAC** = $20,000 / 500 = **$40 per customer**
- ARPU = $10 / month
- Gross Margin = 80%
- Avg. Lifetime = 12 months
- **LTV** = 10×0.8×12 = **$96 per customer**

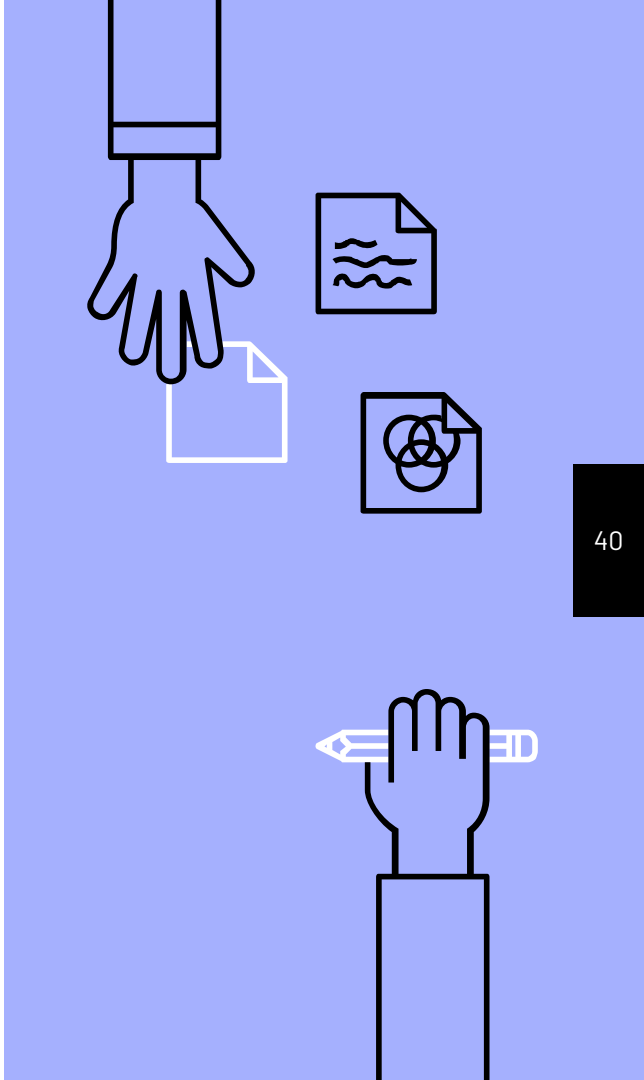
- **Ratio LTV : CAC** = 96/40 = **2.4**

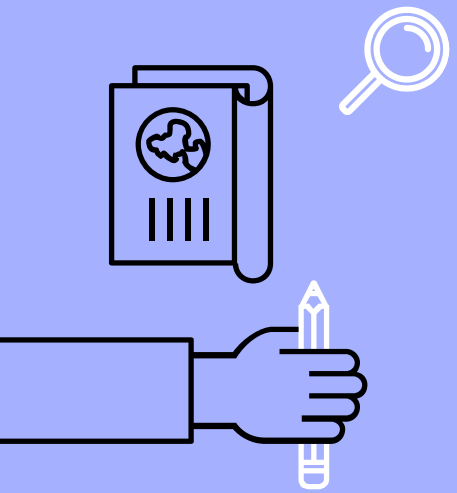


LTV : CAC RATIO

- A healthy business typically maintains **LTV ≥ 3 × CAC**
- If **LTV:CAC < 1**, you're losing money on each user.

Would you rather have a low CAC or a high LTV — and why?

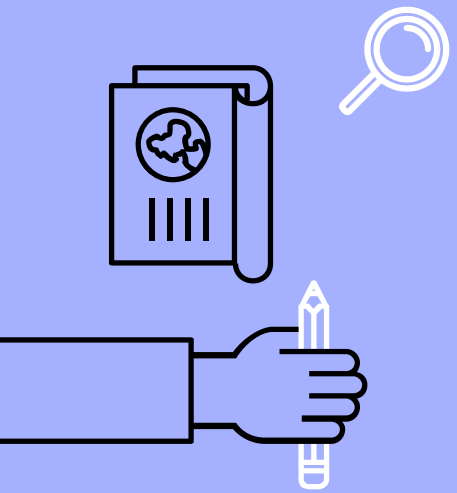




Fix the Unit Economics

Which lever to improve first — CAC, Retention, or Pricing?

	Startup A	Startup B	Startup C
CAC	60€	25€	100€
ARPU	12€/month	8€/month	25€/month
Churn	10%	20%	5%
LTV	120€	40€	500€
LTV : CAC	2 : 1	1.6 : 1	5 : 1

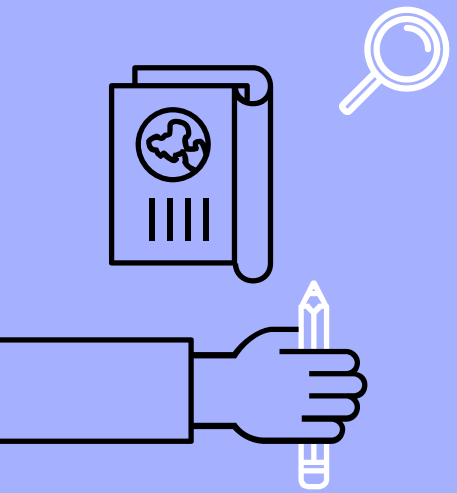


Correction: Startup A

A leaky bucket: fix retention first

Today: LTV 120€ · LTV:CAC 2:1 · CAC payback 5 months		
Lever	**To reach LTV : CAC = 3**	**Verdict**
CAC	60€ → 40€ (–33%)	Hard: paid CAC rises as you scale
Retention	Churn 10% → 6.7%	**Best pick: 1st lever, it compounds the active base**
Pricing	ARPU 12€ → 18€ (+50%)	Risky: a price shock raises churn

LTV = ARPU / monthly churn (gross margin = 100 %, as in the quizzes). Target: LTV : CAC ≥ 3.

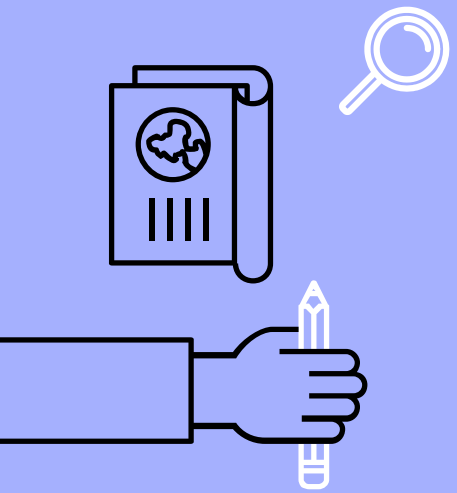


Correction: Startup B

No product-market fit yet: stop scaling, fix the product

Today: LTV 40€ · LTV:CAC 1.6:1 · users stay 5 months on average		
Lever	**To reach LTV : CAC = 3**	**Verdict**
CAC	25€ → 13€ (–47%)	Unrealistic, and CAC is already low
Retention	Churn 20% → 10.7%	**Best pick: half the users leave every 3 months**
Pricing	ARPU 8€ → 15€ (+88%)	Unrealistic before users see more value

LTV = ARPU / monthly churn (gross margin = 100 %, as in the quizzes). Target: LTV : CAC ≥ 3.



Correction: Startup C

Healthy, maybe too cautious: invest in growth

Today: LTV 500€ · LTV:CAC 5:1 · CAC payback 4 months		
Lever	**To reach LTV : CAC = 3**	**Verdict**
CAC	Can rise to 167€ and stay at 3:1	**Best pick: buy more growth, even at a higher CAC**
Retention	Churn already 5% (20-month lifetime)	Protect it: monitor cohorts as you scale
Pricing	Test a premium tier	Upside, second priority

LTV = ARPU / monthly churn (gross margin = 100 %, as in the quizzes). Target: LTV : CAC ≥ 3.

Churn & Retention

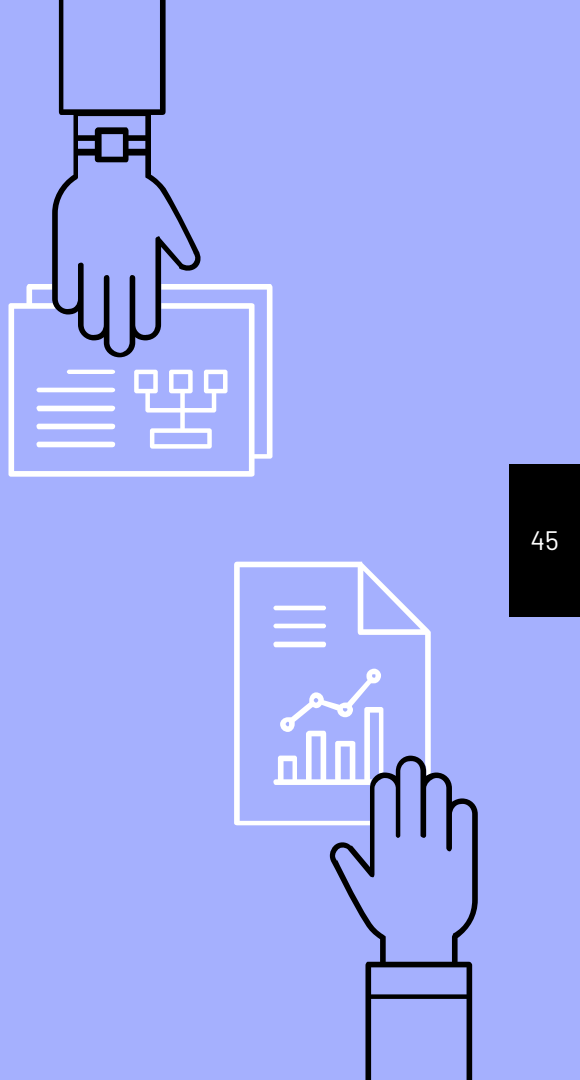
$$Churn = \frac{\text{Customers at Start} - \text{Customers at End}}{\text{Customers at Start}}$$

Retention = 1 - Churn

- Example:
  - Start: 1,000 customers
  - End: 850 customers

→ Churn = (1000 - 850) / 1000 = 15%

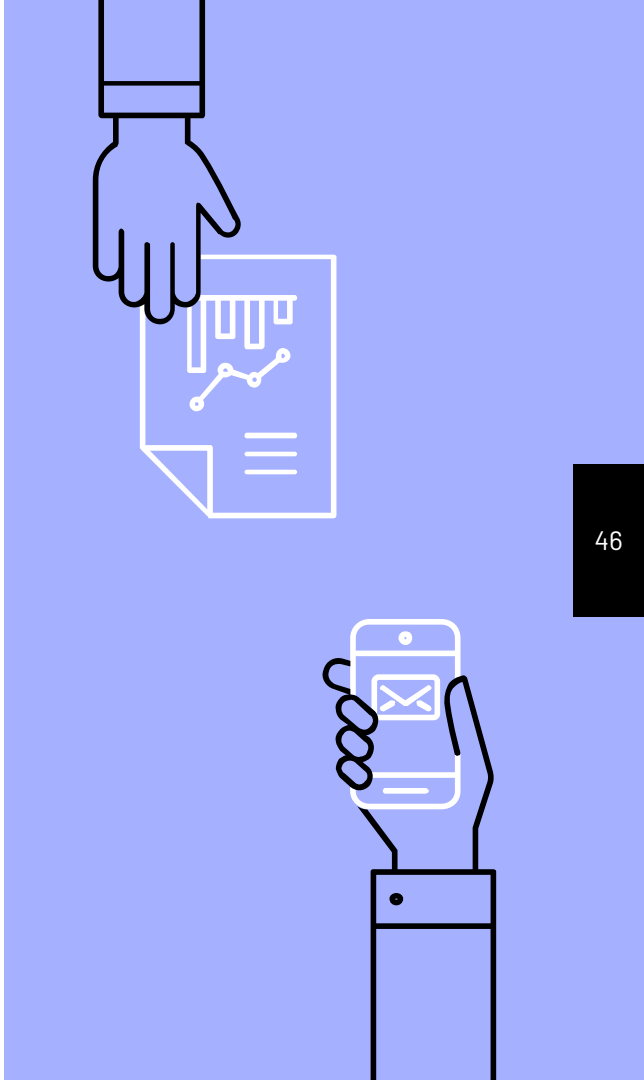
→ Retention = 85%



COHORT ANALYSIS

Group users by signup date (cohort) and track retention over time.

Signup month	Month 1	Month 2	Month 3	Month 4
January	100%	80%	60%	45%
February	100%	85%	70%	50%
March	100%	90%	75%	60%

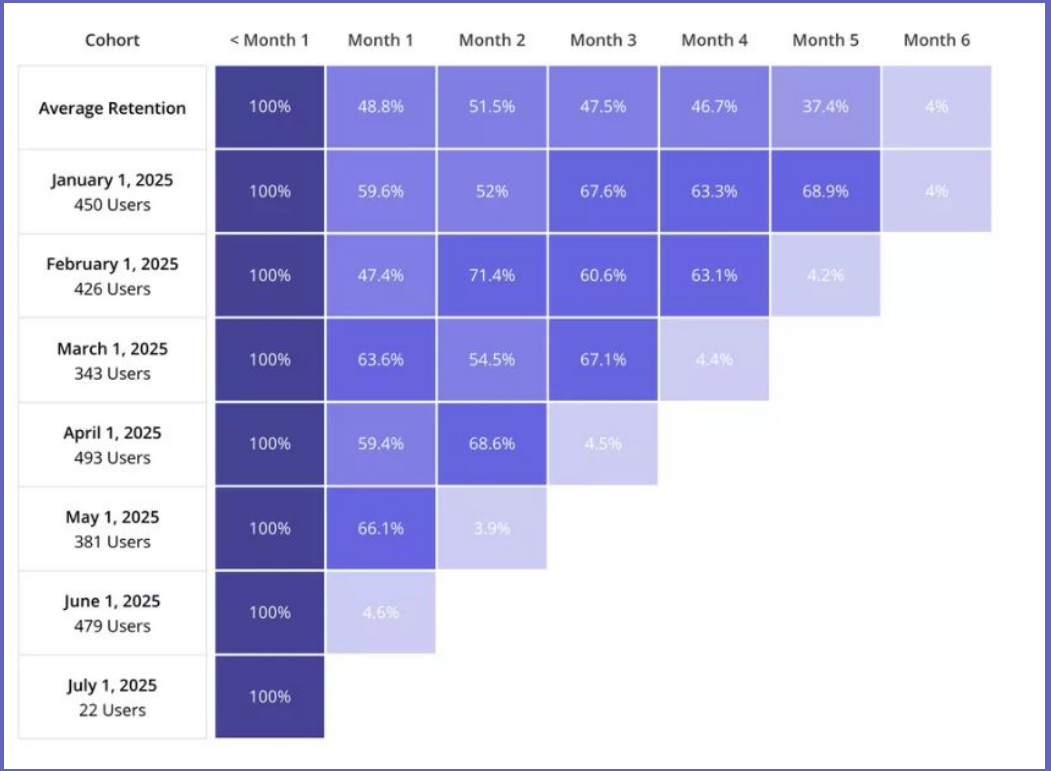


Retention improving over time

→ product improvements or better onboarding.

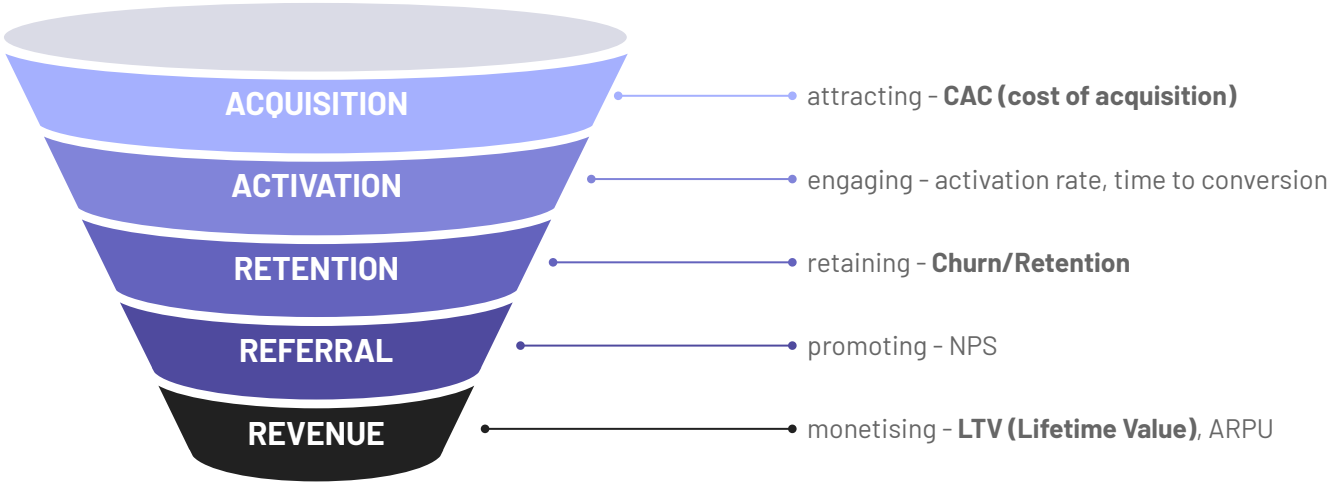
Retention declining

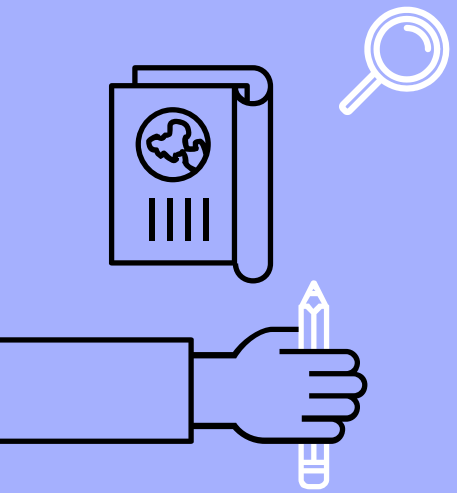
→ user fatigue or poor engagement.



Cohort	< Month 1	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Average Retention	100%	48.8%	51.5%	47.5%	46.7%	37.4%	4%
January 1, 2025 450 Users	100%	59.6%	52%	67.6%	63.3%	68.9%	4%
February 1, 2025 426 Users	100%	47.4%	71.4%	60.6%	63.1%	4.2%	
March 1, 2025 343 Users	100%	63.6%	54.5%	67.1%	4.4%		
April 1, 2025 493 Users	100%	59.4%	68.6%	4.5%			
May 1, 2025 381 Users	100%	66.1%	3.9%				
June 1, 2025 479 Users	100%	4.6%					
July 1, 2025 22 Users	100%						

FUNNEL AARRR

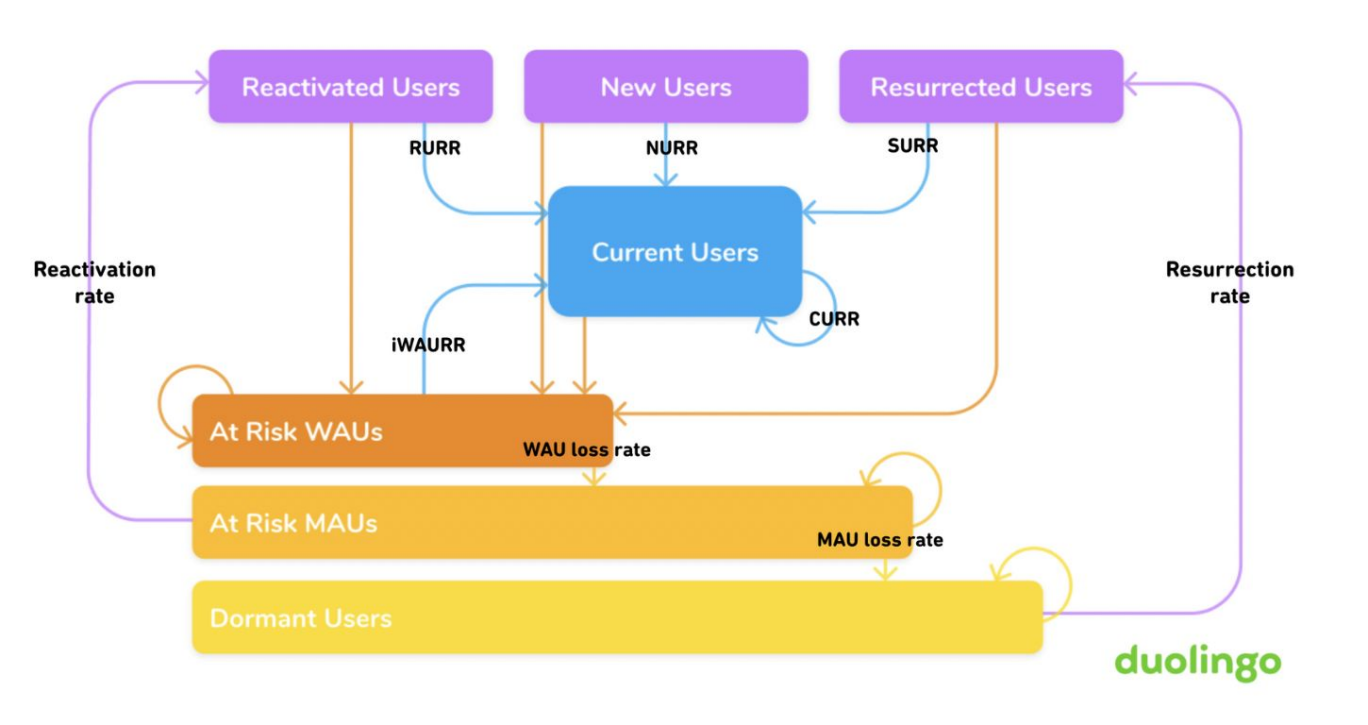




Cohort exercise

- You start with 1,000 users.
- Monthly churn at 10%.
- What happens if you reduce churn by just 2 percentage points (from 10% → 8%)?
- How many more users stay after 6 months?

DUOLINGO EXAMPLE



Virality & Word of Mouth

How many new users each existing user brings in?

$$K = i \times c$$

- **i** = invites per user
- **c** = conversion rate

- Example :
  - 1000 users
  - each user invites 2 friends → i = 2
  - 30 % of those friends sign up → c = 0.3
  - Then: K = 2 × 0.3 = 0.6

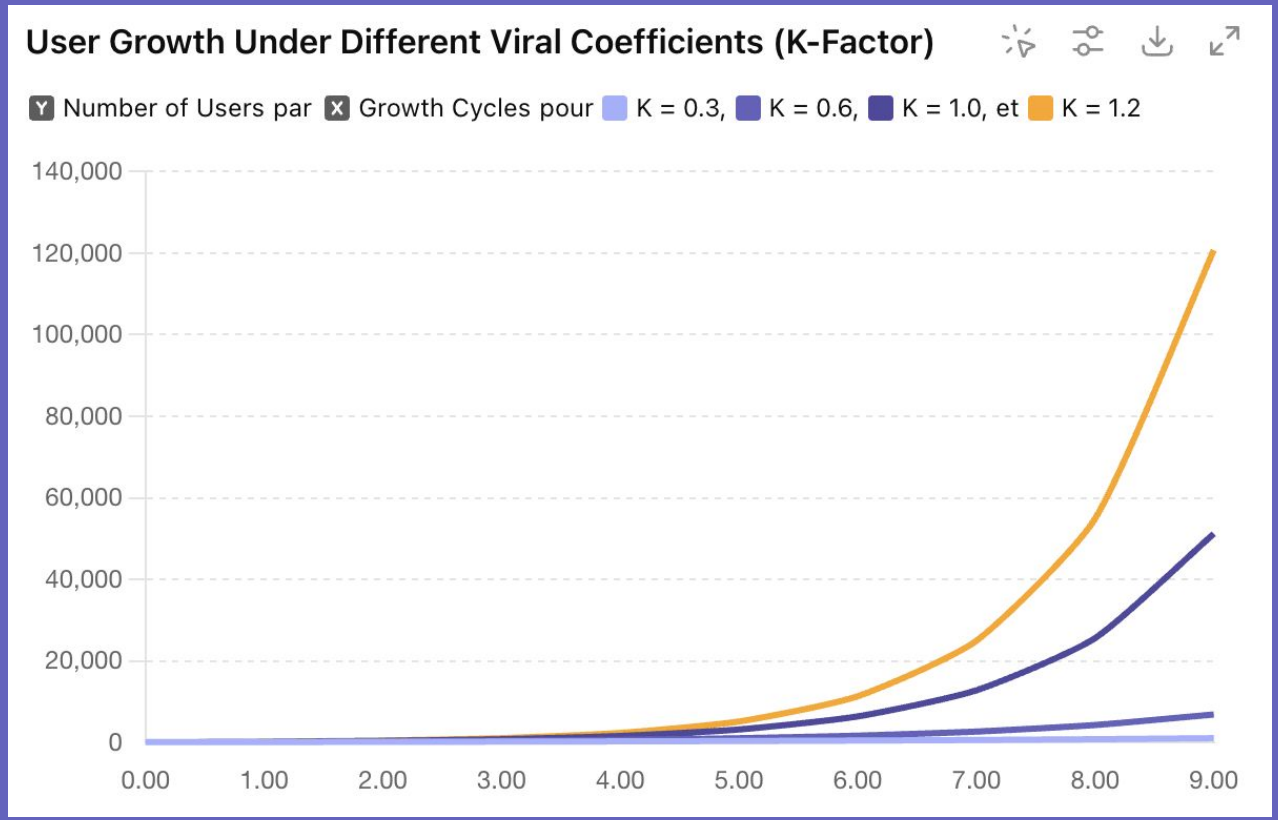
→ every user generates 0.6 new users on average



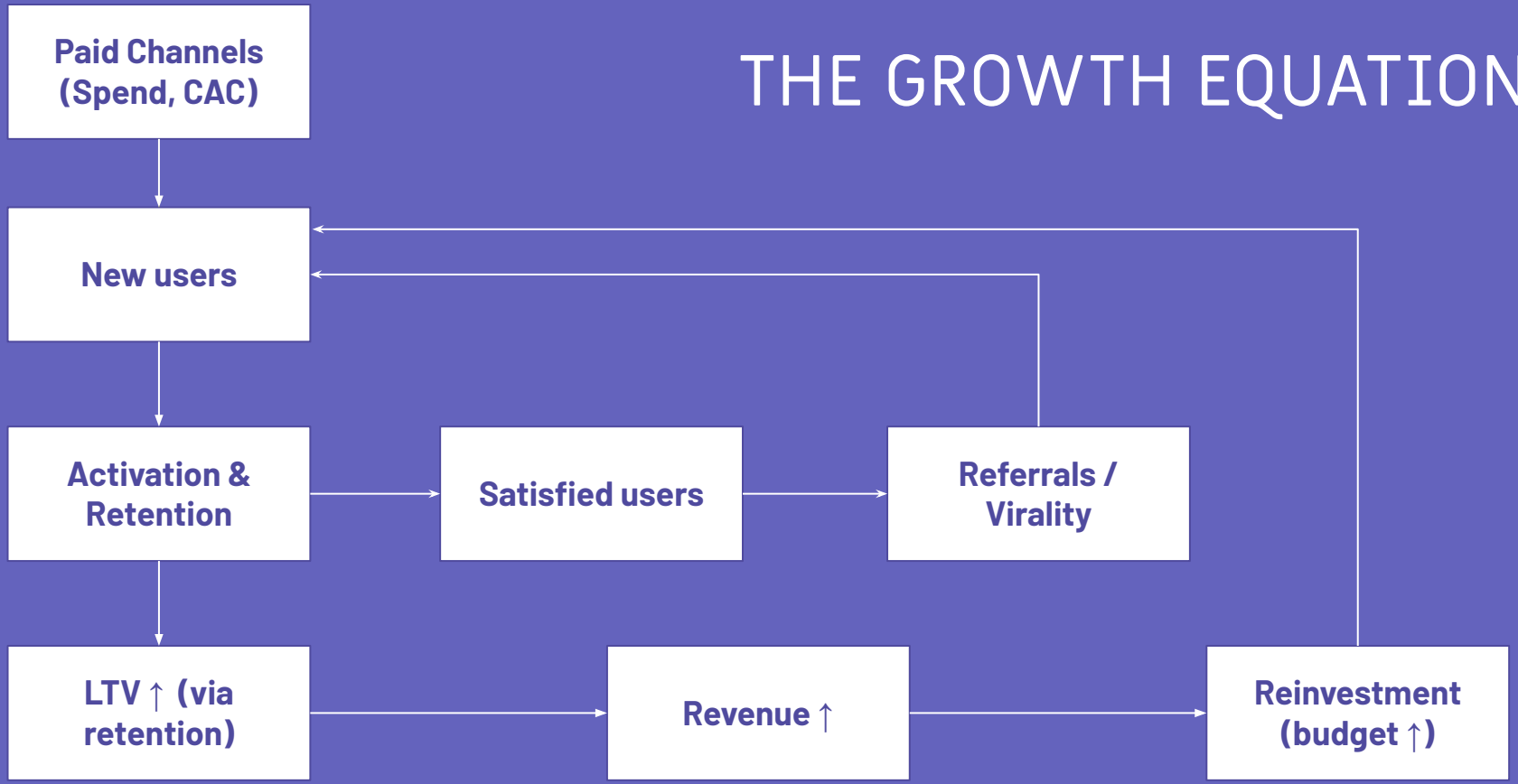
Virality & Word of Mouth

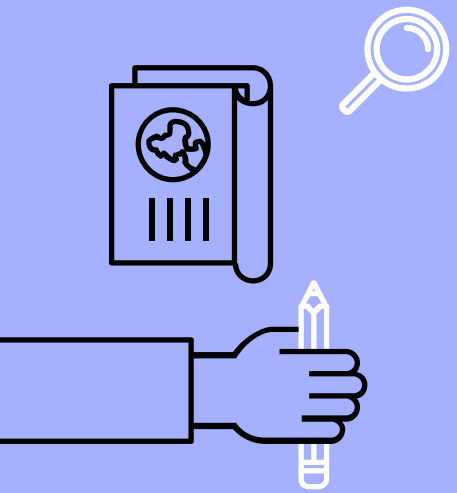
- **K > 1** → the loop fuels itself (viral growth engine).
- **K < 1** → the loop helps, but doesn't replace acquisition.
- **K ≈ 0.3–0.8** → realistic sweet spot; good products use it to stretch their marketing efficiency, not eliminate it. Not viral alone but multiplies ads efficiency.




User Growth Under Different Viral Coefficients (K-Factor)
Y Number of Users par
X Growth Cycles pour
K = 0.3,
K = 0.6,
K = 1.0, et
K = 1.2
140,000
120,000
100,000
80,000
60,000
40,000
20,000
0
0.00
1.00
2.00
3.00
4.00
5.00
6.00
7.00
8.00
9.00

THE GROWTH EQUATION



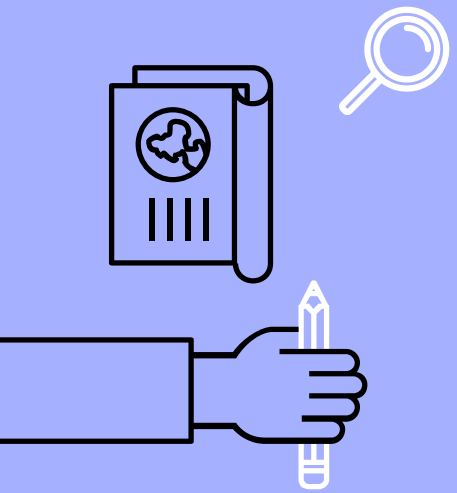


Time for a quiz (1/3)

A startup spends €40 000 to acquire 1 000 new paying users. Each user pays €20/month and churns at 10 % per month.

Which statement is true?

A. CAC = €20; LTV ≈ €200 → exceptional (10:1)
B. CAC = €40; LTV ≈ €200 → healthy (5:1)
C. CAC = €40; LTV ≈ €100 → unsustainable
D. CAC = €20; LTV ≈ €100 → strong growth loop

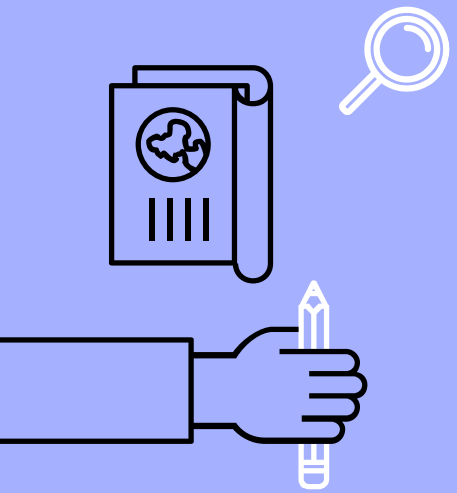


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Time for a quiz (2/3)

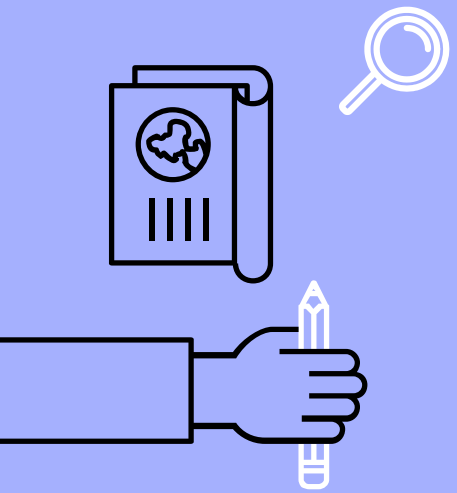
If a product improves its monthly churn from 10 % to 5 %, what happens to its LTV, assuming ARPU = €10?

A. LTV stays the same — churn doesn't affect value.

B. LTV decreases — fewer users churn means less new acquisition.

C. LTV increases from €100 to €150.

D. LTV doubles from €100 to €200.



Time for a quiz (2/3)

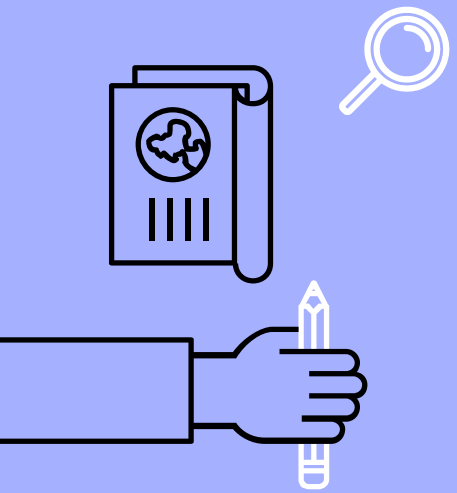
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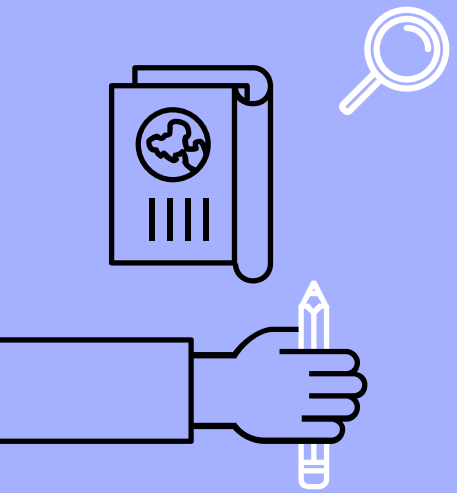
D. LTV doubles from €100 to €200.



Time for a quiz (3/3)

An app's users send an average of **2 invites** each, and **25 %** of invitees join. What's the app's viral coefficient (K-factor), and will it grow exponentially?

A. K = 0.25 ⟶ no, it'll decay
B. K = 0.5 ⟶ no, it needs fuel
C. K = 1 ⟶ yes, it's self-sustaining
D. K = 2 ⟶ yes, infinite growth



Time for a quiz (3/3)

An app's users send an average of **2 invites** each, and **25 %** of invitees join. What's the app's viral coefficient (K-factor), and will it grow exponentially?

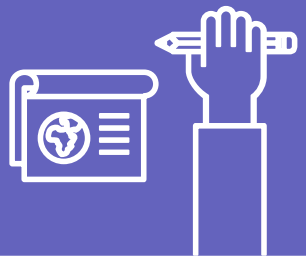
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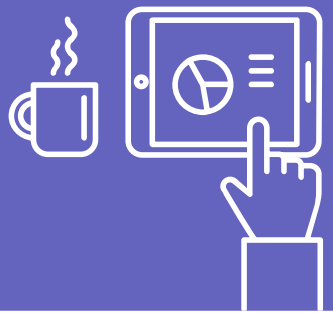
Which metric is most dangerous to ignore?

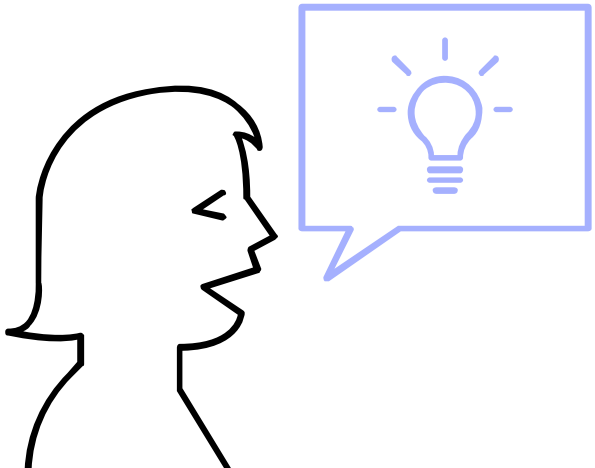


Why can a low CAC be a bad sign?

How does retention signal product market fit?







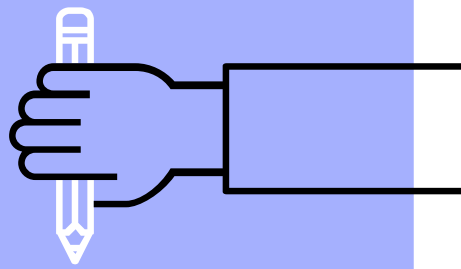
CAC:LTV ratio determines scalability

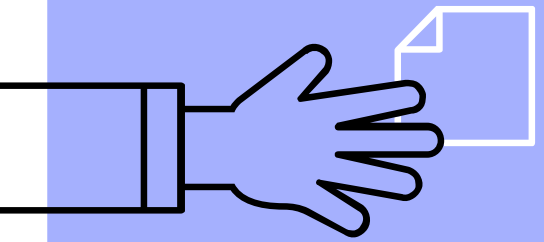
Retention compounds faster than acquisition.

Virality amplifies retention.

Improving retention is usually the smartest growth lever.

3.
ACQUISITION & THE ETHICS OF GROWTH





Not all growth is created equal.

Paid Channels

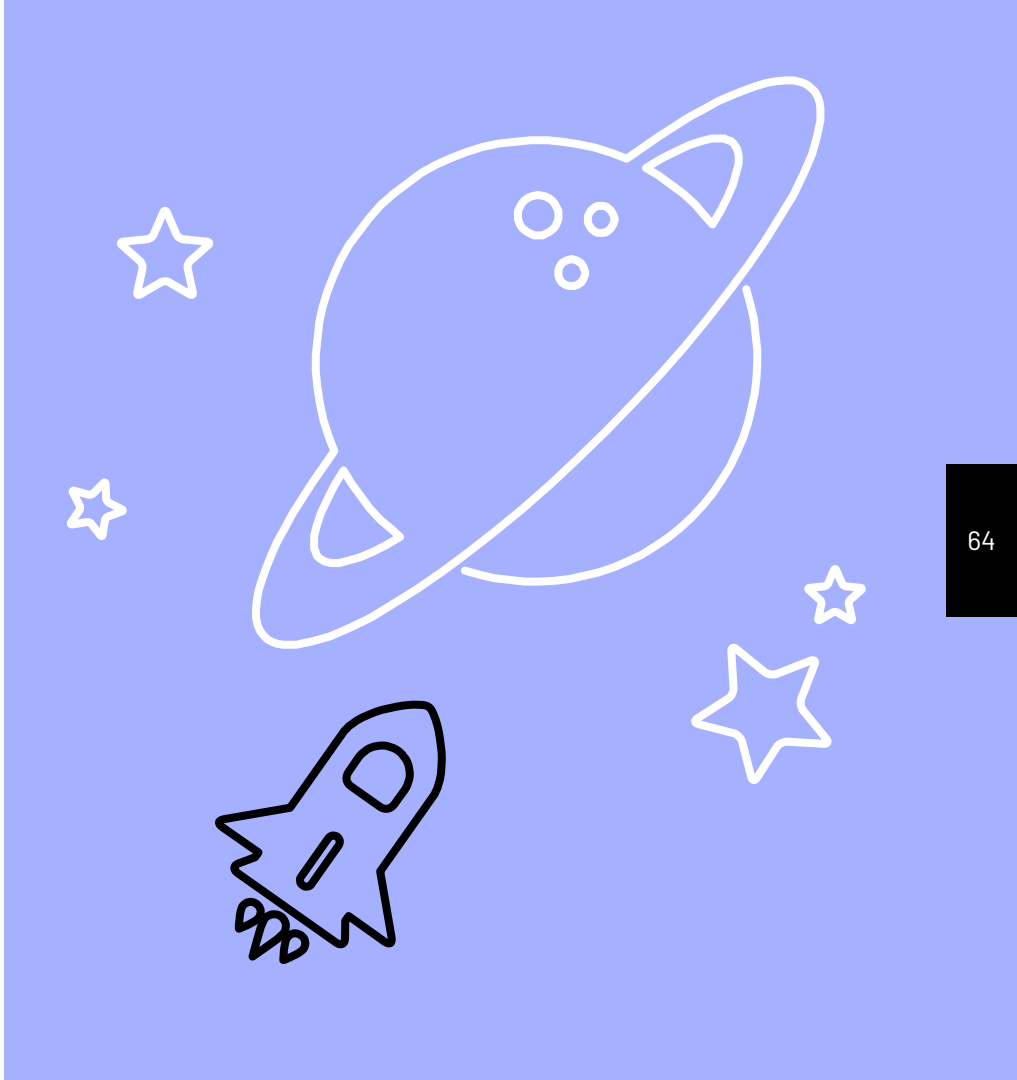
Fast but costly

Organic Channels

Slow but durable

Referrals & Incentives

Dark Patterns & Ethics



WHERE DID YOU HEAR ABOUT IT?

Think of the last 3 apps or products you started using.

How did you discover them?



“

Growth

isn’t one channel;

it’s a portfolio.



THE ACQUISITION SPECTRUM

Paid	Organic
• Ads • Influencers • Affiliates	• SEO • Community • Referrals

PAID CHANNELS: THE FAST LANE

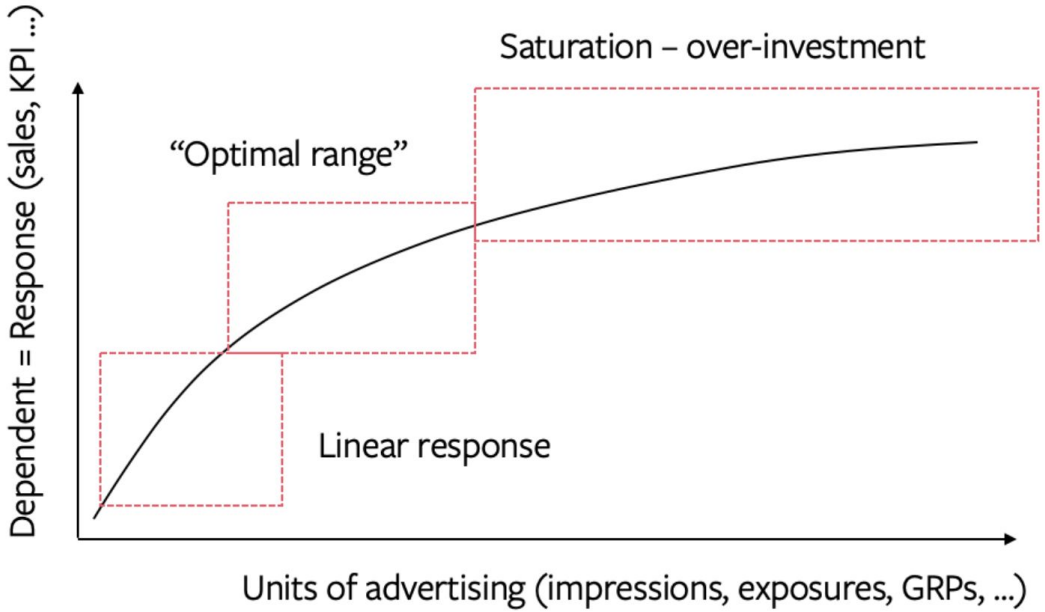
Paying for attention

- **Search** (Google Ads) - Intent driven
- **Social** (Meta, TikTok) - Interest driven
- **Display** - Interest driven
- Cost metrics: CPM, CPC, CPA

$$CAC = \frac{CPC}{ConversionRate}$$

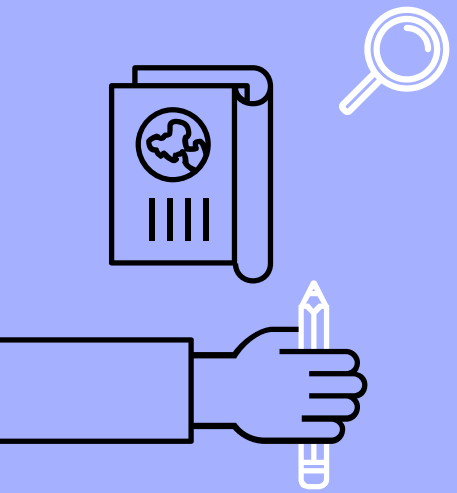


PAID CHANNELS DYNAMICS



After a certain point, also called the saturation point, each dollar you spend has a relatively lower impact.

" Why does doubling ad budget often not double users?"



Paid Channel exercise

- Startup spends €20k/month on ads
- CAC = €50
- churn = 10%
- You get +10 % budget

- Do you spend, optimize, or pivot?

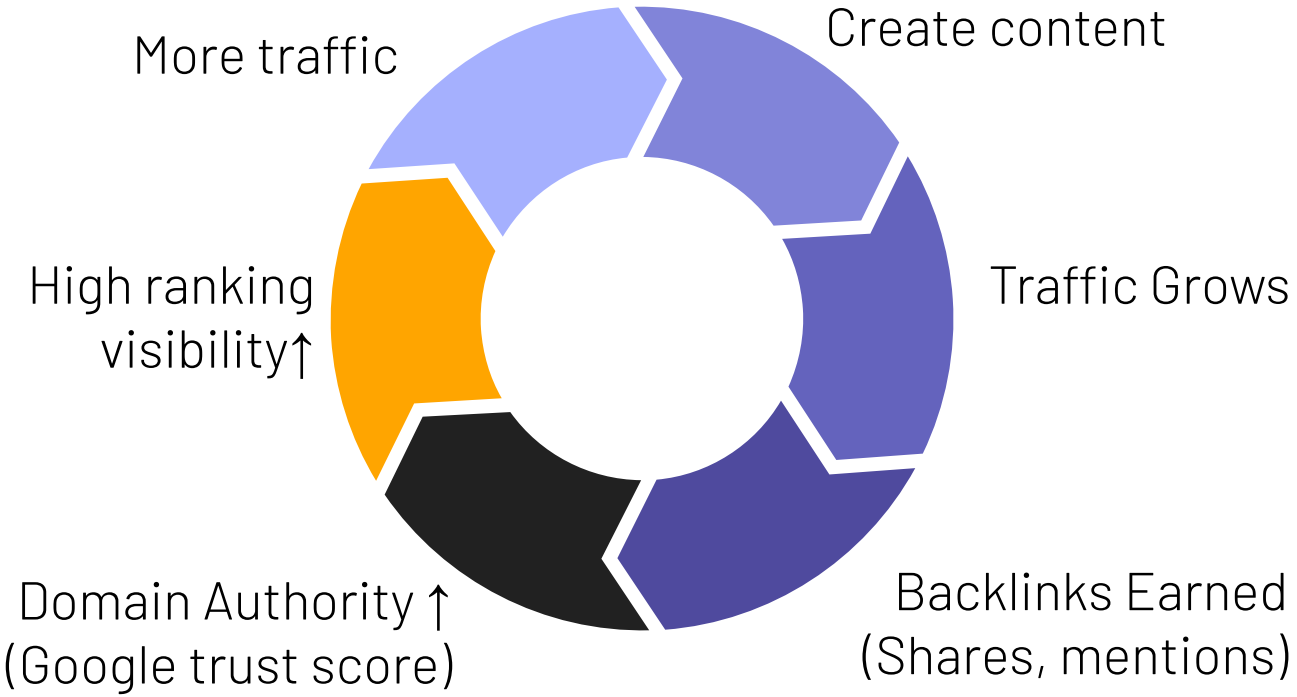
ORGANIC CHANNELS: THE LONG GAME

Building trust

- SEO
  - search-based, high intent, long setup time.
- Content Marketing
  - builds authority and brand.
- Community & Product-Led Growth
  - advocacy, word of mouth through product value.
- Referrals & Virality:
  - built-in acquisition loops.

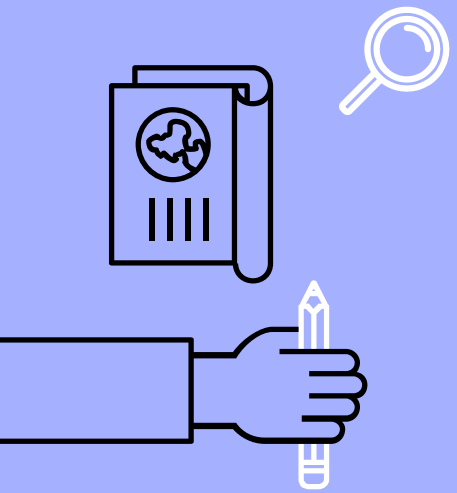


THE SEO FLYWHEEL



PAID vs ORGANIC TRADEOFFS

	🕑 Fast Impact	⏳ Slow Impact
💸 **Paid**	**Ads, Influencers, Affiliates** - Instant reach - Scalable with budget - Diminishing returns (rising CAC)	**Partnerships, Sponsorships** - Brand alignment builds credibility - Slow setup, but durable reach
🌱 **Organic**	**Referrals, Social Virality** - Unpredictable spikes - Hard to control, low cost - Strong trust signal	**SEO, Content, Community** - Compounding traffic - High upfront effort, near-zero CAC long term



Paid Channel exercise

Imagine you're launching a new mobility app. You have €100 000 marketing budget and 6 months to grow active users.

You can split your effort between:

1️⃣ Paid channels (ads, influencers, sponsorships)

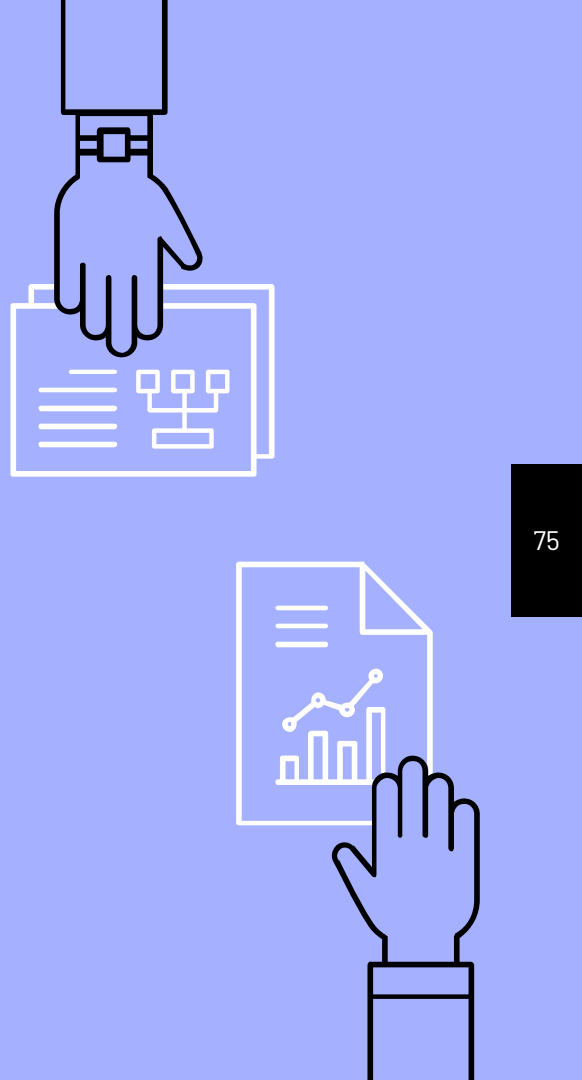
2️⃣ Organic channels (SEO, community, referrals, product virality)

- How would you allocate the budget — and why? Think about speed, cost, risk, and long-term payoff.

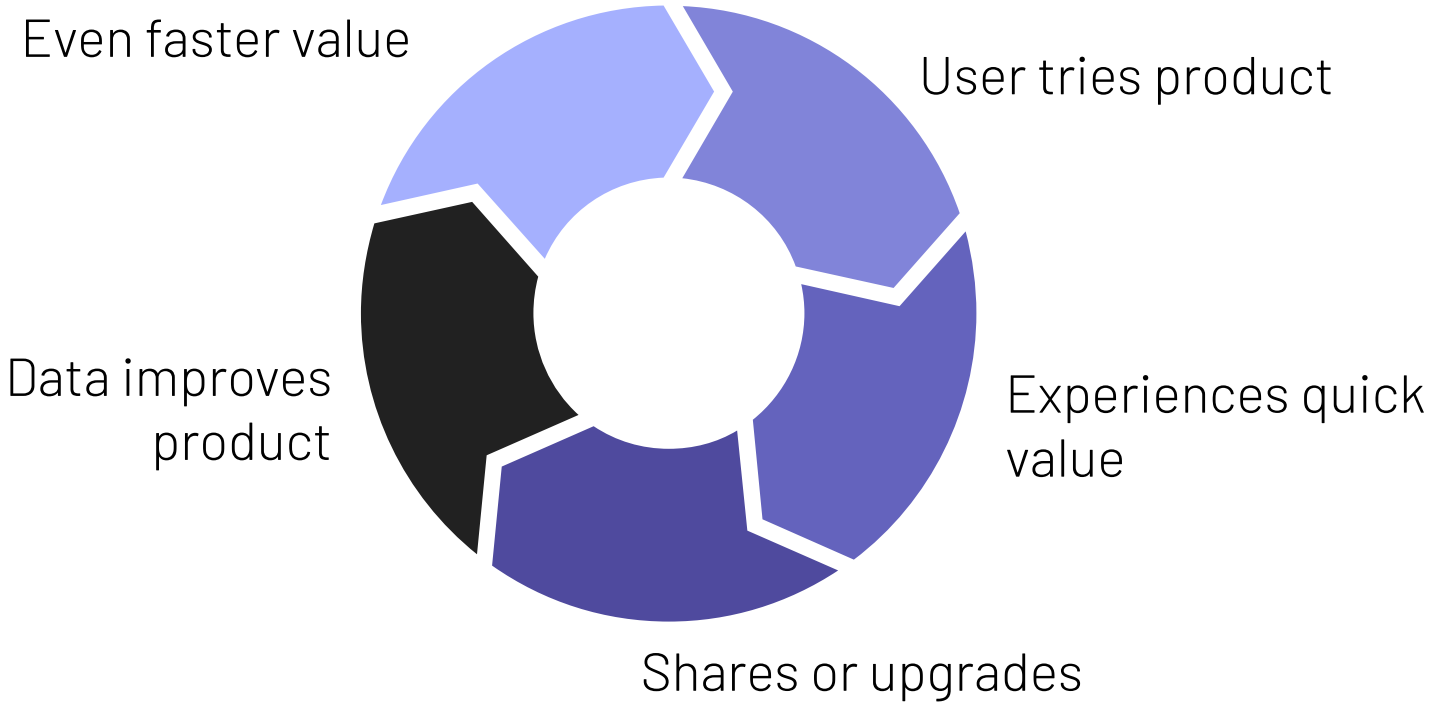
PRODUCT LED GROWTH

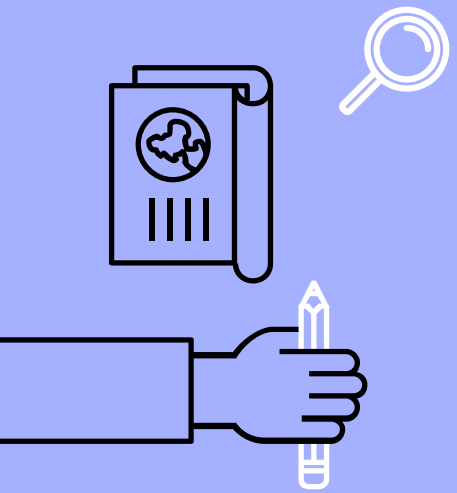
A go-to-market strategy where the product itself drives acquisition, activation, and retention.

- **The product = the main growth engine**
  Instead of relying primarily on sales or marketing
- Users **experience value**
  they convert, invite others, and expand usage.
- The **feedback loop** between product usage and business metrics is immediate and measurable



THE PRODUCT FLYWHEEL





PRODUCT EXERCISE

- Find a company that has a strong product
- Analyse the product and identify how its design leads to growth
- 3 bullet points (e.g. Slack, Notion, Figma)

REFERRALS & INCENTIVES

Self-reinforcing growth mechanism where each new user can directly create more users — if the incentive and experience align

- **The mechanics:**

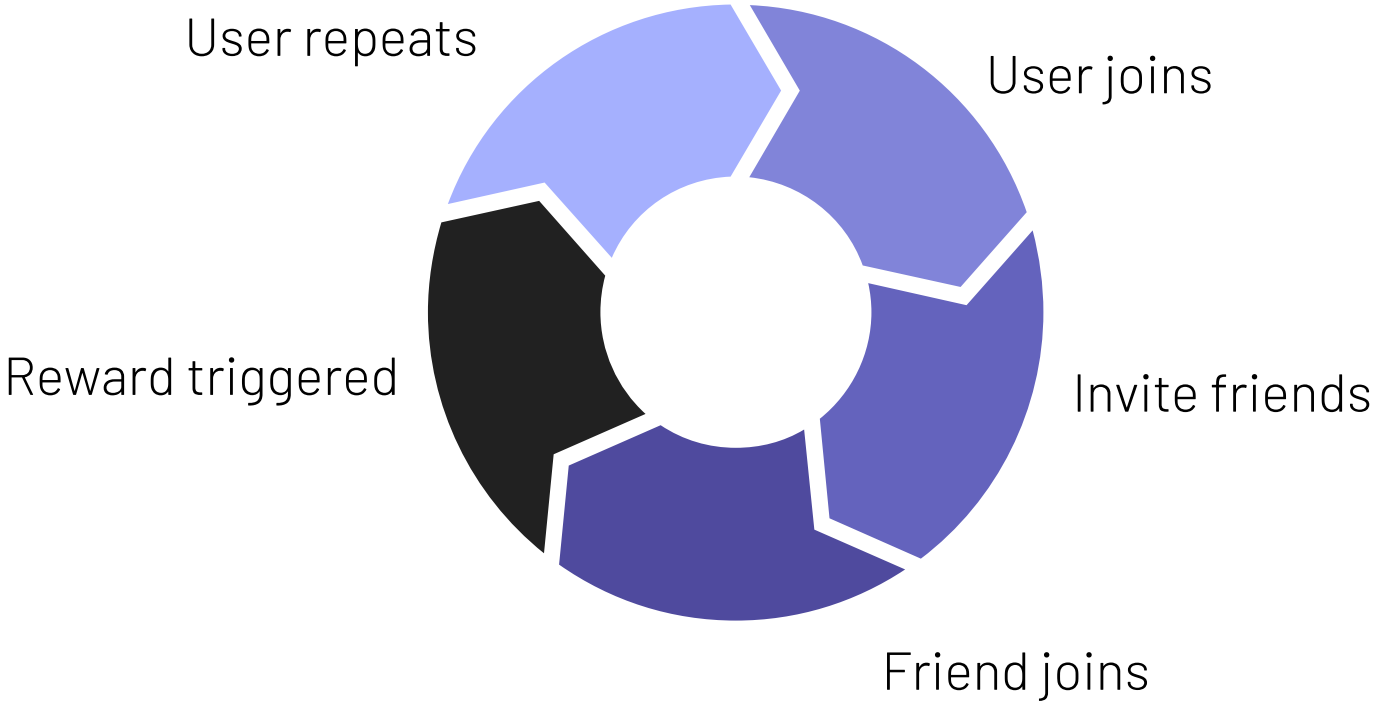
  Trigger → Incentive → Simplicity.
- **Behavioral loop**

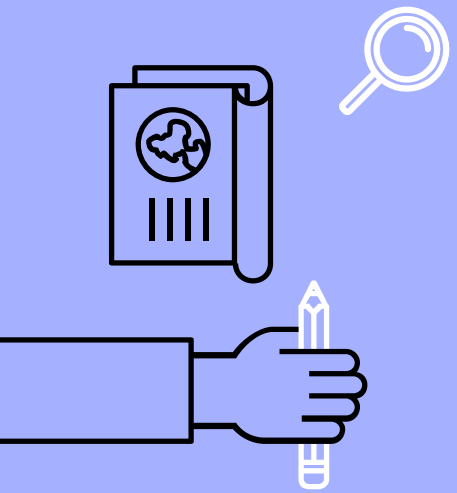
  that depends on motivation, trust, and design.
- Viral coefficient K = i x c
- The **reward structure** determines how likely users are to refer

  ⚠️ Pitfall: over-incentivization → low-quality users.



THE REFERRAL FLYWHEEL





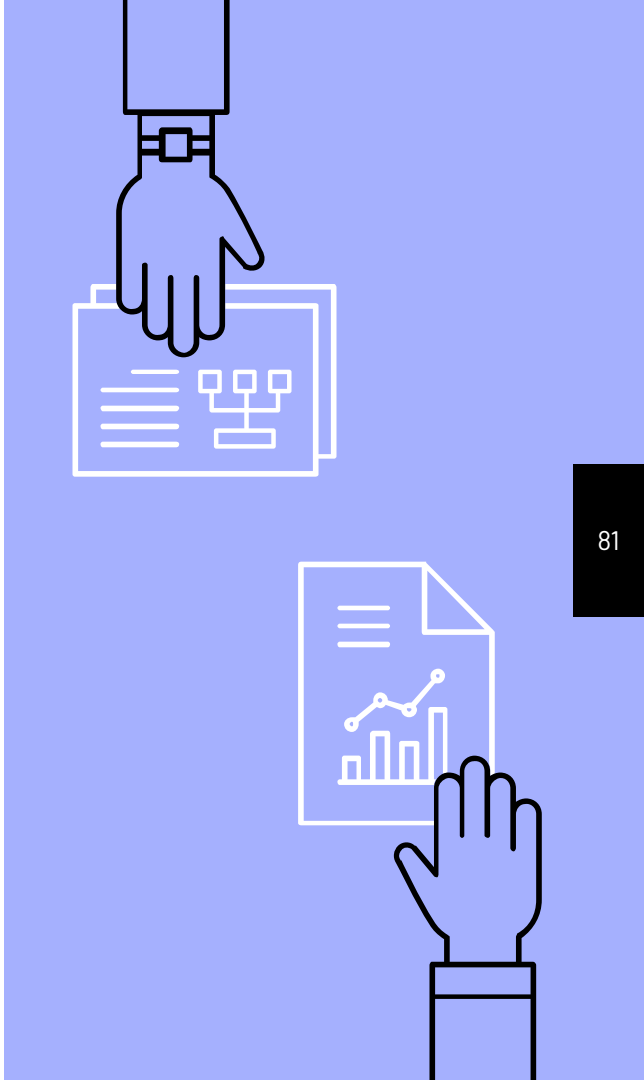
REFERRAL EXERCISE

- You’re designing a referral loop for an online fashion marketplace. You can reward users with:
  - €10 cash
  - Store credit or discount
  - Exclusive early access to new collections
- Which do you choose, and why?

DARK PATTERNS

When growth crosses the line

- Hidden unsubscribe buttons
- Guilt-shaming copy (“Are you sure you want to miss this deal?”)
- Auto-opt-ins / forced sharing

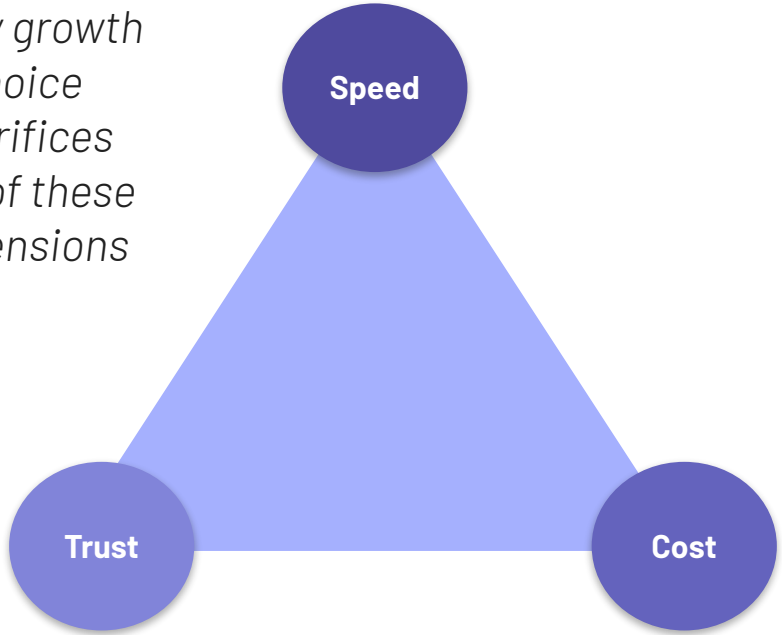


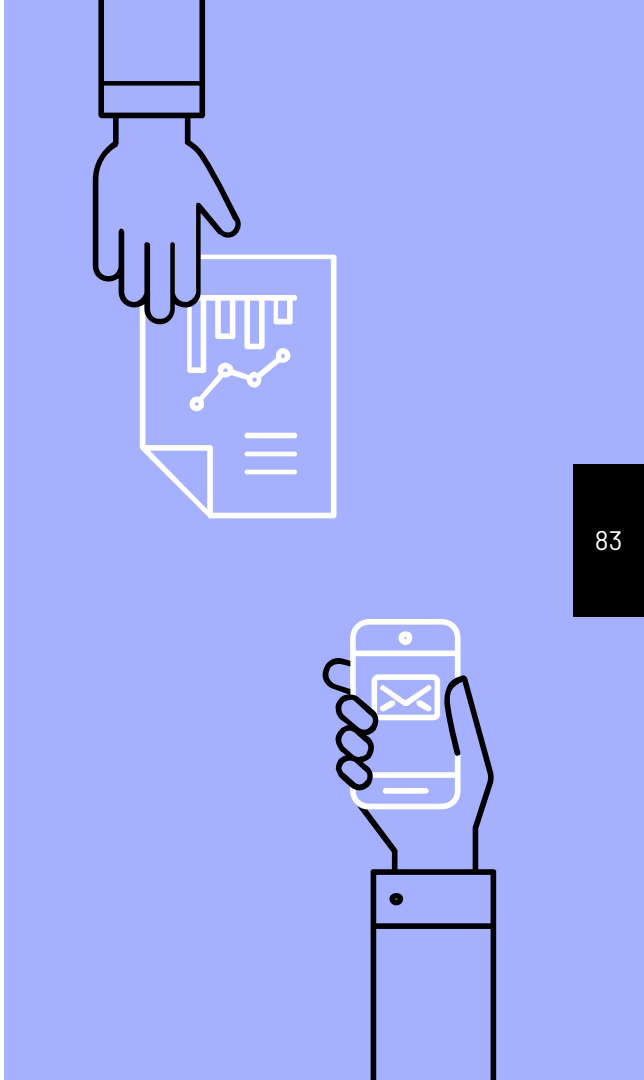
DARK PATTERNS EXAMPLES

Company	Pattern Type	Short-Term Win	Long-Term Risk
LinkedIn	Forced virality	Fast user growth	Lawsuits, user backlash
Facebook	Data opacity	Higher engagement	Regulatory scrutiny
Amazon	Obstruction	Lower churn	Loss of brand goodwill
Instagram	Addictive loop	High retention	Mental health & PR issues
Duolingo	Emotional triggers	Daily actives ↑	Ethical criticism
Ryanair	Hidden costs	Higher ARPU	Customer distrust

BALANCING GROWTH FORCES

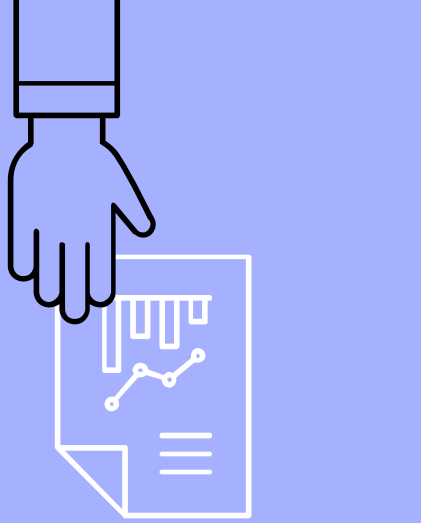
Every growth choice sacrifices one of these dimensions





EXAMPLES

Growth Type	Strength	What you sacrifice	Strategic Use
Paid	Speed	Cost & Trust	Early acceleration, testing, launching
Organic	Trust	Speed & Control	Sustainable base, brand moat
Referral / Virality	Balance (efficiency + Credibility)	Predictability & Control	Mid-stage multiplier, network effects



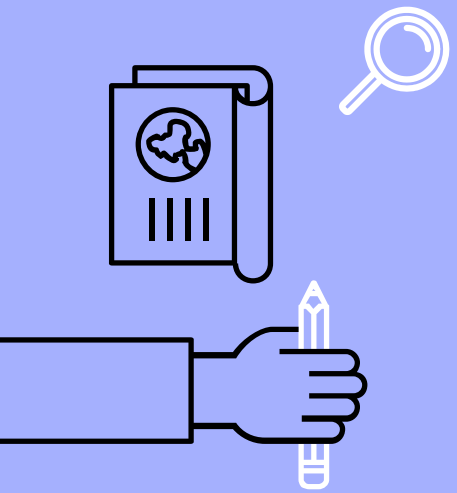


Ethical Growth Principles

Sustainable growth is not anti-growth — it's long-term optimization

- Align incentives with **user value**
- **Be transparent** about trade-offs
- Optimize for **lifetime trust**, not one-time clicks



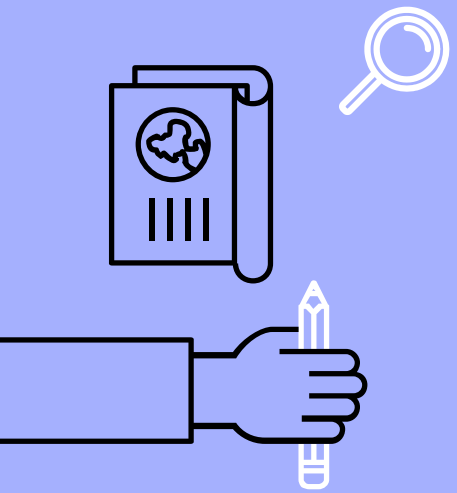


Time for a quiz (1/3)

A new app claims a “K-factor” of 1.2 because each user sends 4 invites and 30% convert. After launch, they discover most new users come from paid ads, not invites.

What’s the real issue?

A. They confused potential virality with effective virality.
B. Their CAC is too high.
C. They need more incentive per invite.
D. Their K-factor formula is wrong.



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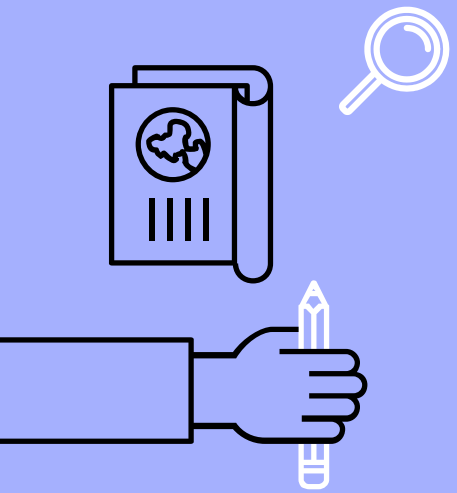
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A. They confused potential virality with effective virality.

B. Their CAC is too high.

C. They need more incentive per invite.

D. Their K-factor formula is wrong.

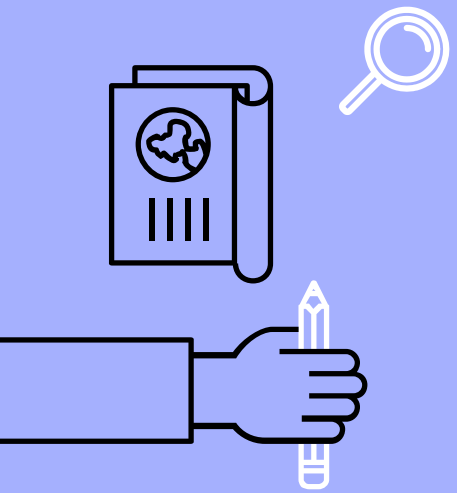


Time for a quiz (2/3)

You run growth for a subscription app. Paid ads deliver users at €40 CAC; SEO delivers at €15 CAC but grows slowly.

What's your best strategic move?

A. Double down on ads for faster growth.
B. Pause SEO — it's too slow.
C. Balance: fund ads to keep growth while compounding SEO for long-term CAC efficiency.
D. Ignore both and launch a referral program.

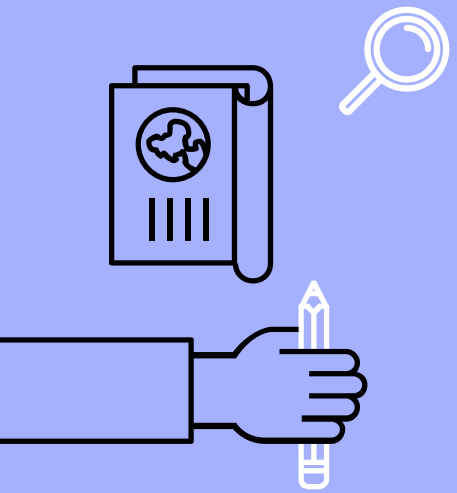


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Time for a quiz (3/3)

A mobile game increases retention by adding surprise pop-ups that exploit “fear of missing out.” Daily actives rise 30%, but user complaints explode.

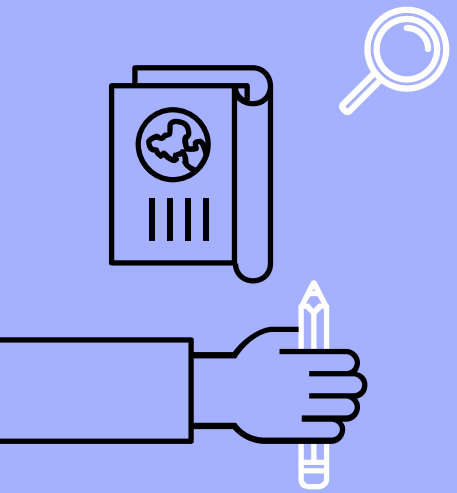
What’s happening?

A. Smart growth: retention up = success.

B. Temporary gain: dark pattern damages trust and long-term retention.

C. Irrelevant - engagement is the only KPI.

D. Needs better A/B testing.



Time for a quiz (3/3)

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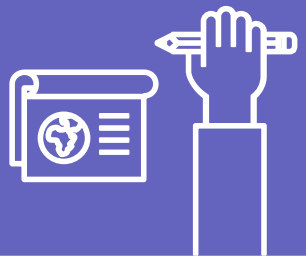
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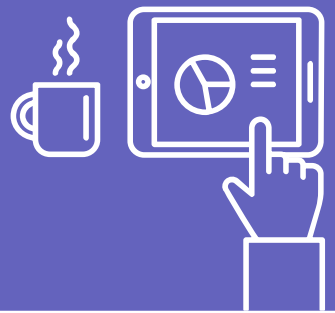
Which channel builds the strongest moat?

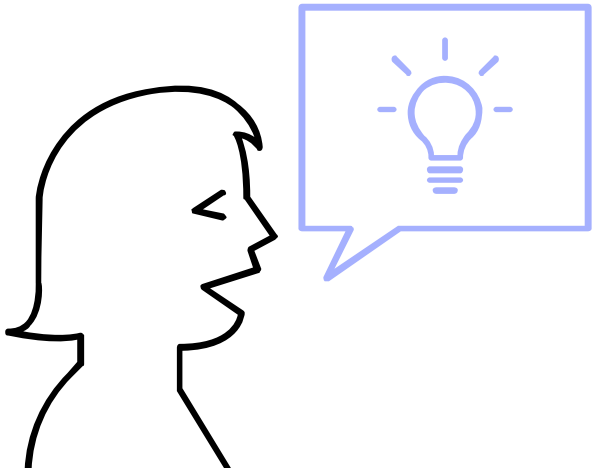


Can growth be 100 % ethical?

Where would you draw the moral line if you ran TikTok's growth team?

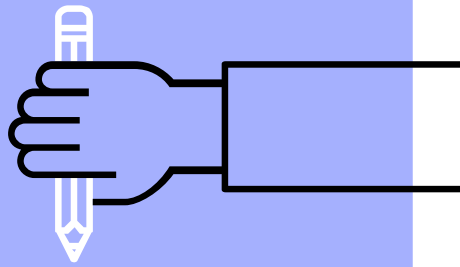


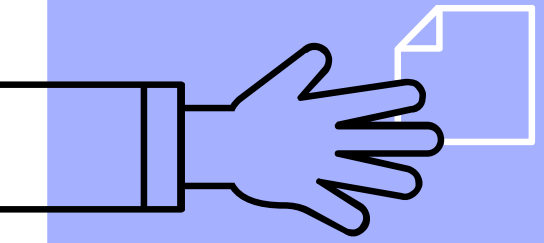




- Paid fuels speed, organic builds the moat
- Referrals & product loops scale trust : user become your best acquisition channel
- Dark patterns break compounding

4.
MODELING GROWTH & SCENARIO BUILDING





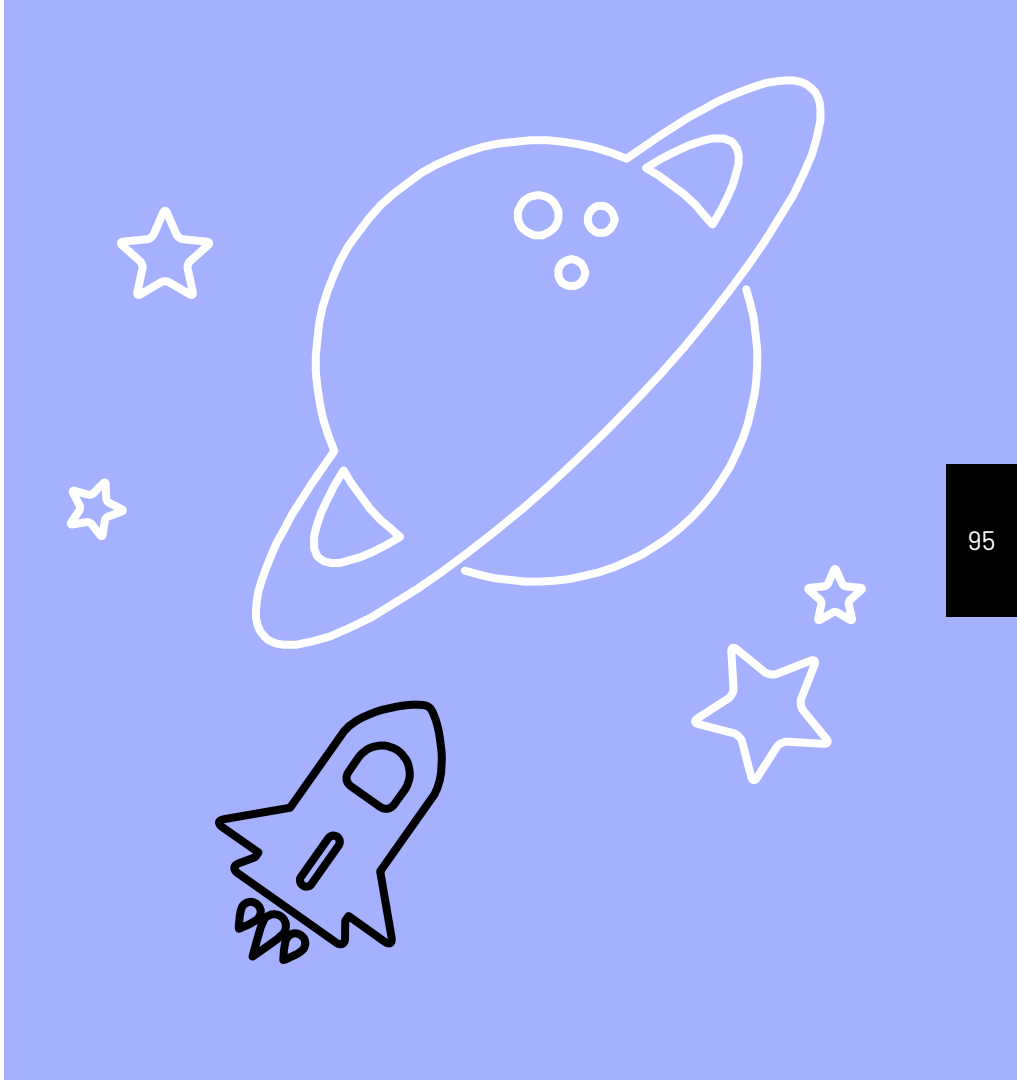
Turning growth from gut feeling into simulation

Anatomy of a Growth Model

Scenario Planning (Optimistic / Base / Pessimistic)

From Numbers → Strategy

Ethics & Realism



IF YOU HAD 10% MORE BUDGET



you're a Growth Lead at a subscription app

- **10 k users**
- **CAC : €40**
- **ARPU : €12/mo**
- **Churn : 8 %**

Where do you invest?



Why model?

Good growth models don't predict the future - they show how fragile your beliefs are.



💰 B2B SAAS GROWTH MODEL

💡 Adjust the Actuals using the months that best represent the baseline for the upcoming quarter.

	Actuals			Simulation		
	August 2024	September 2024	October 2024	M+1	M+2	M+3
GROSS NEW $ARR	**$327,220**	**$428,334**	**$330,059**	**$361,381**	**$361,381**	**$361,381**
TRIAL $ARR	**$238,211**	**$264,410**	**$219,319**	**$241,421**	**$241,421**	**$241,421**
Trial forms submitted	**1000**	**980**	**1111**	1030	1030	1030
Verified trials	**850**	**843**	**900**	**865**	**865**	**865**
Trial verification rate	85.00%	86.00%	81.00%	84.00%	84.00%	84.00%
New accounts created	**770**	**774**	**811**	**786**	**786**	**786**
Account creation rate	77.00%	79.00%	73.00%	76.33%	76.33%	76.33%
New customers (PT)	**147**	**149**	**160**	**152**	**152**	**152**
Trial conversion rate	14.70%	15.20%	14.40%	14.77%	14.77%	14.77%

ANATOMY OF A GROWTH MODEL



- Spend
- CAC
- Users Acquired

- Activation Rate
- Churn
- Virality
- ARPU

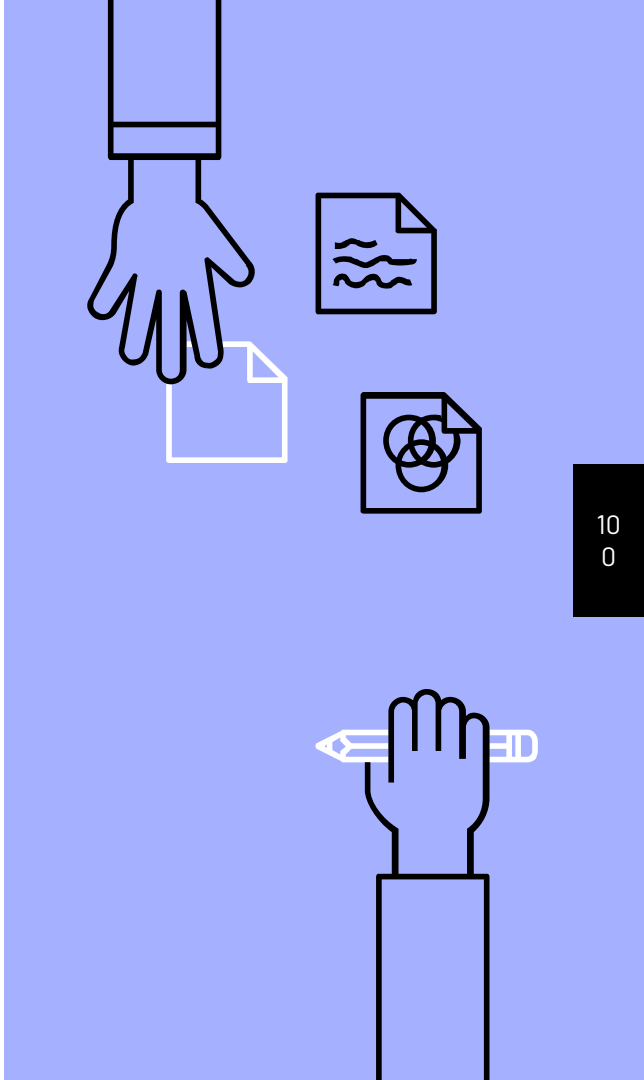
- Revenue
- Active Users
- Growth Rate

$$U_{t+1} = (U_t \times (1 - Churn)) + (NewUsers) + (Referrals)$$

EXAMPLE

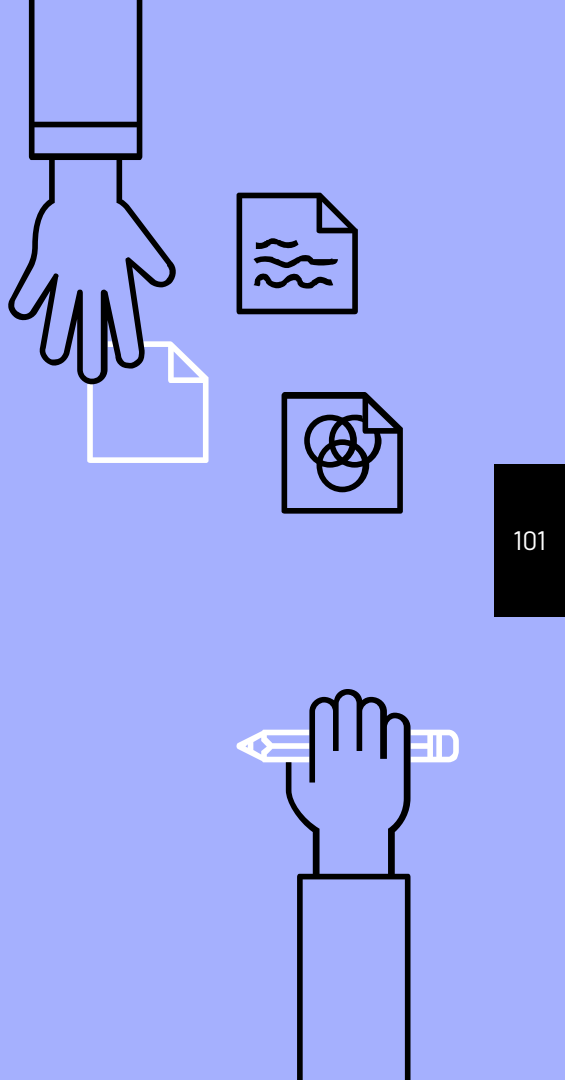
- Start with: **U_0 = 1000 users**
- CAC = €50 → Spend €5000 → 100 new users
- Churn = 10 % → 900 remain
- Virality = 0.05 → 45 referrals
- **U1 = 1 045 next month** → +4.5 % growth

- What if churn drops to 5 %?
  - 1098
- Virality = 0.1?
  - 1090

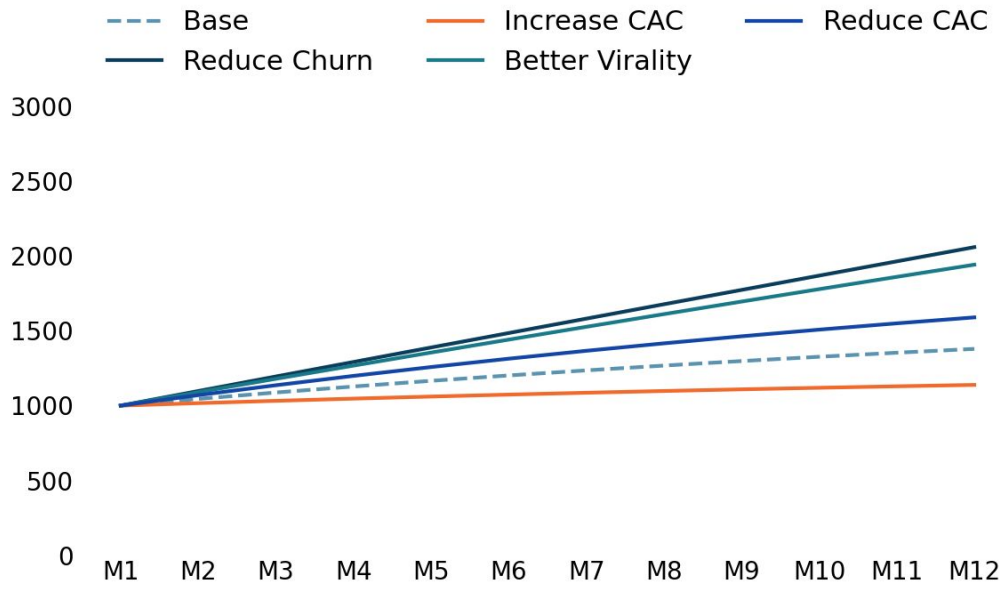


WHAT IF EXPERIMENTS

- Test scenarios as controlled “what if” experiments :
  - **Base case**: current realistic parameters.
  - **Optimistic**: best plausible improvements (virality ↑, Retention ↑).
  - **Pessimistic**: worst-case shocks (e.g., CAC ↑, churn ↑).



WHAT IF EXPERIMENTS



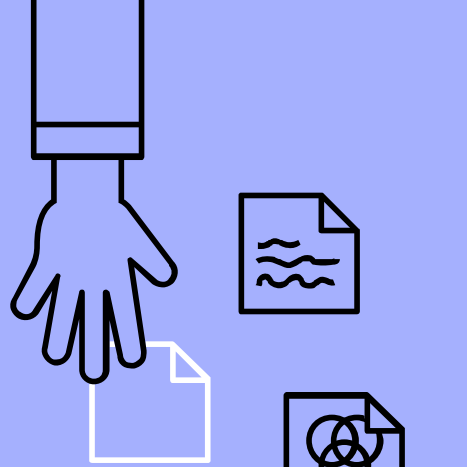
We tweak one parameter at a time

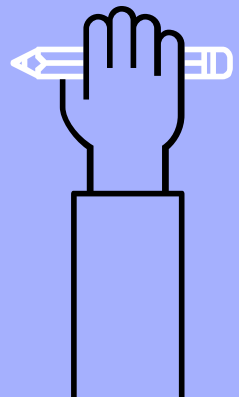
- **churn from 10% to 5%**
- **CAC from 50€ to 70€**
- **CAC from 50€ to 40€**
- **Virality from 0.05 to 0.1**

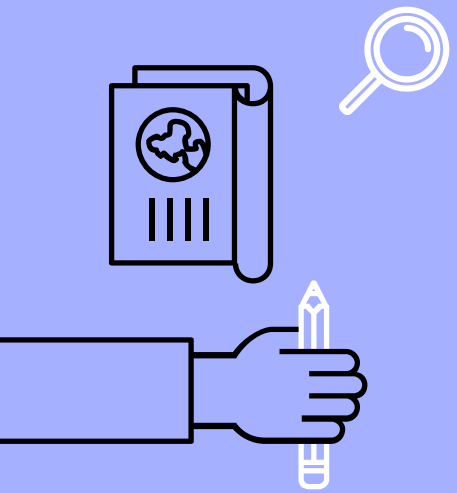
FROM NUMBERS TO STRATEGY

Once you model outcomes, you can derive strategic priorities

Metric	Strategic Lever	Example Action
CAC ↑	Channel Mix	Shift to organic or community
Retention ↓	Product / UX	Improve activation flow
ARPU ↓	Monetization	Upsells or pricing review
K ↓	Network design	Referral simplification







MODELING EXERCISE

Imagine you are the VP of Growth for a B2B SaaS product currently acquiring 1,000 new sign-ups per month. You have an extra $10,000 budget this quarter, and your goal is to maximize Monthly Active Users (MAU) 6 months from now."

You have two strategic options for investment. Which one yields a higher MAU growth, and why?

Scenario 1: Spend $10,000 on product improvements (UX, onboarding).

Scenario 2: Spend $10,000 on paid ads.

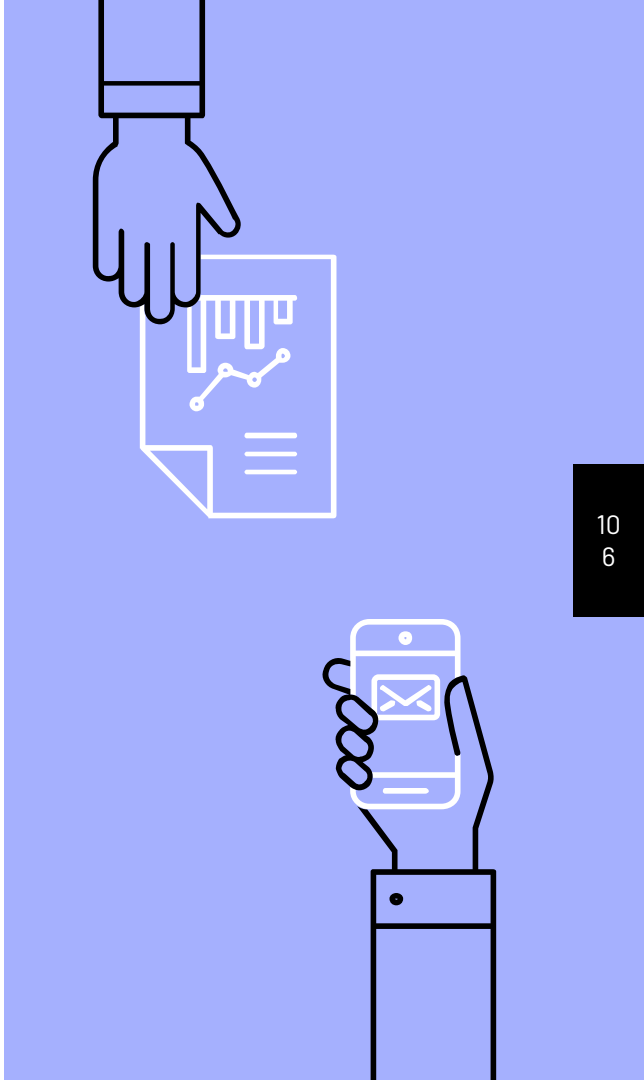
ETHICS & REALISM

If your best scenario depends on doubling screen time, are you creating value or addiction?

MODELING PITFALLS

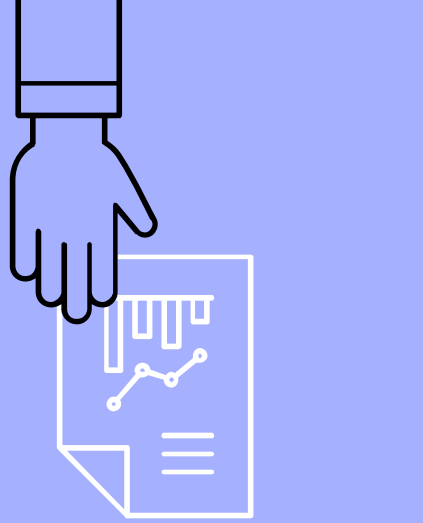
⚠️ Common errors:

- Assuming linear growth forever.
- Ignoring cost of retention initiatives.
- Confusing virality with word of mouth.

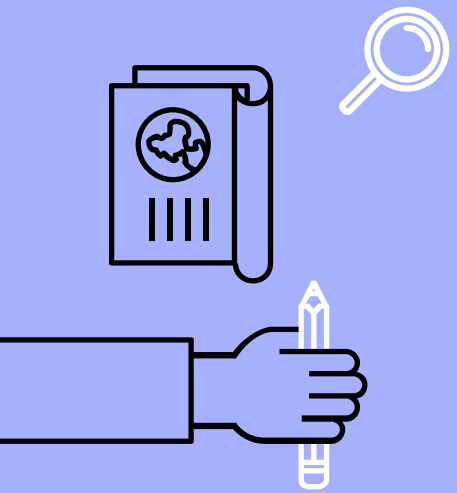


FULL SYSTEM RECAP

Concept	Purpose	Question to ask
Funnel	Diagnose	Where do we leak?
Loop	Sustain	What reinforces itself?
Metrics	Measure	What drives what?
Channels	Execute	Where do we grow?
Model	Integrate	What if we change X?







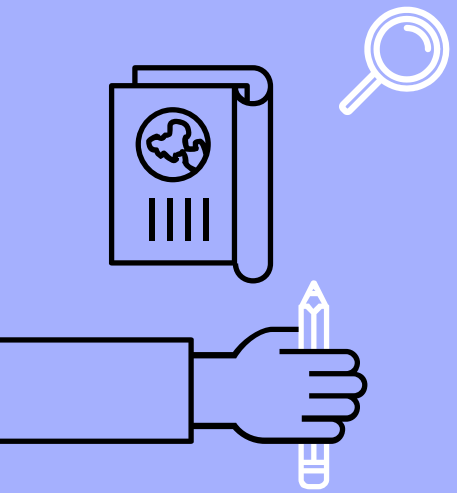
Time for a quiz (1/3)

You build a growth model assuming:

- CAC = €50
- Churn = 5 %
- Monthly ARPU = €25
- Then churn actually rises to 10 %.

What happens to your LTV : CAC ratio?

A. It improves because churn doesn't affect LTV
B. It halves because LTV halves when churn doubles
C. It stays the same because CAC is constant
D. It doubles because higher churn = more frequent purchases



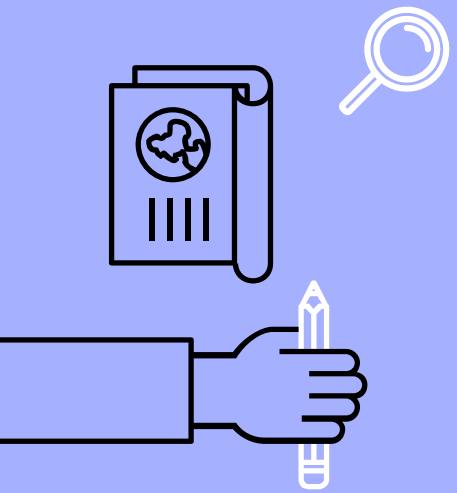
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Time for a quiz (2/3)

Your base model assumes **K = 0.4** (each user brings 0.4 new ones). In an optimistic scenario, **K = 0.8** and churn drops from 8 % → 6 %.

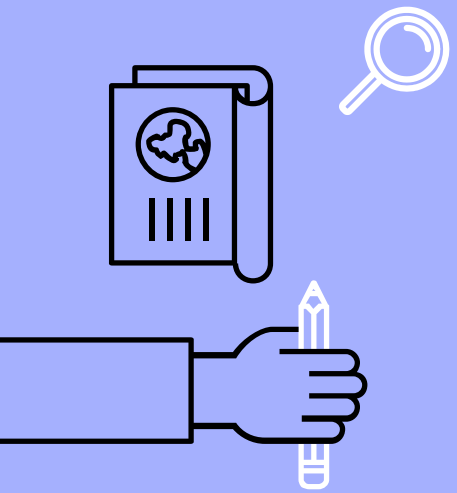
What's the most likely outcome after a few cycles?

A. Growth flattens — virality still < 1

B. Growth accelerates exponentially and becomes self-sustaining

C. Growth doubles instantly in the first cycle

D. Growth slows down because virality reduces CAC



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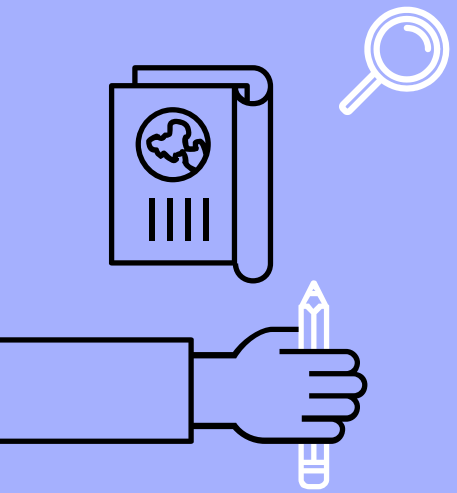
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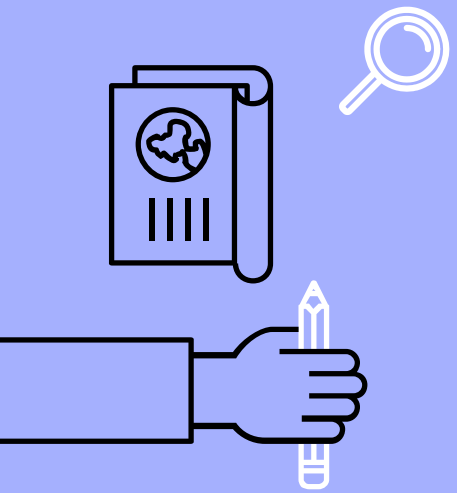
Time for a quiz (3/3)

Your base model:

- CAC = €40
- ARPU = €10/month
- Churn = 10 %

You can improve only one lever by 20 %.
Which gives the biggest impact on LTV : CAC ratio?

A. Reduce CAC by 20 %
B. Increase ARPU by 20 %
C. Reduce churn by 20 % (10 % → 8 %)
D. A and C tie (+25 % each).



Time for a quiz (3/3)

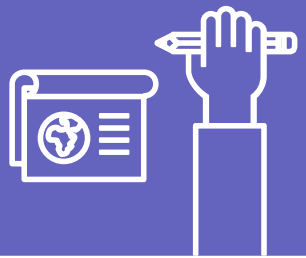
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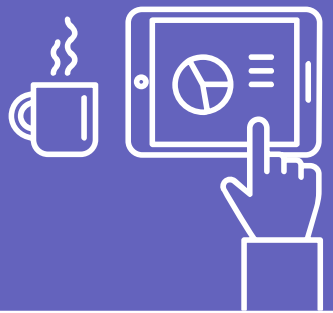
What's the most sensitive assumption in any growth model?

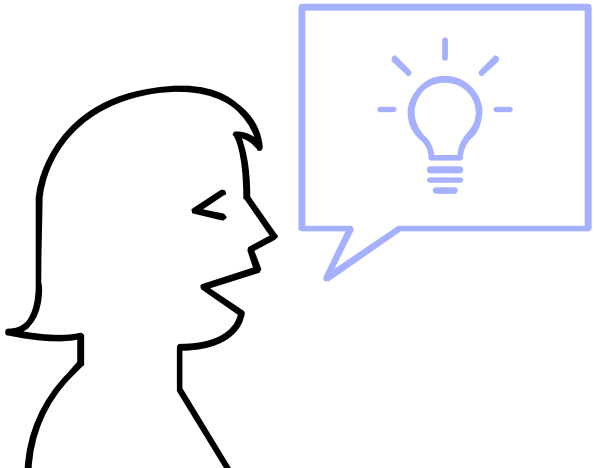




Should models be realistic or aspirational?

Can a growth model be ethical if it ignores impact on users?





- Growth models turn metrics into systems.
- Scenarios reveal risk and potential.
- Retention and virality drive non-linear returns.
- Ethical constraints build resilient companies.



To sum up

Funnels describe,

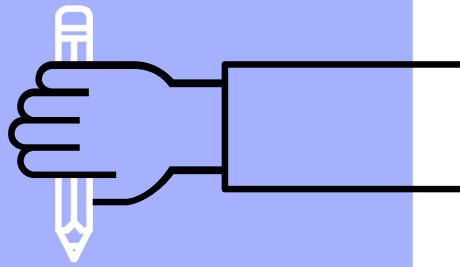
loops drive,

metrics measure,

channels execute,

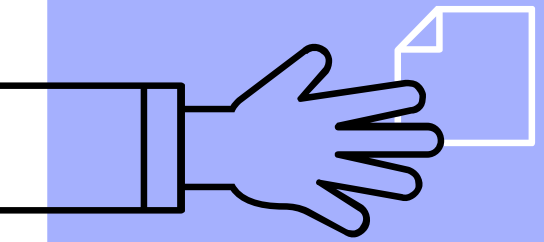
models integrate





5.
THE PROJECTS

Now let’s practice !



YOUR MISSION

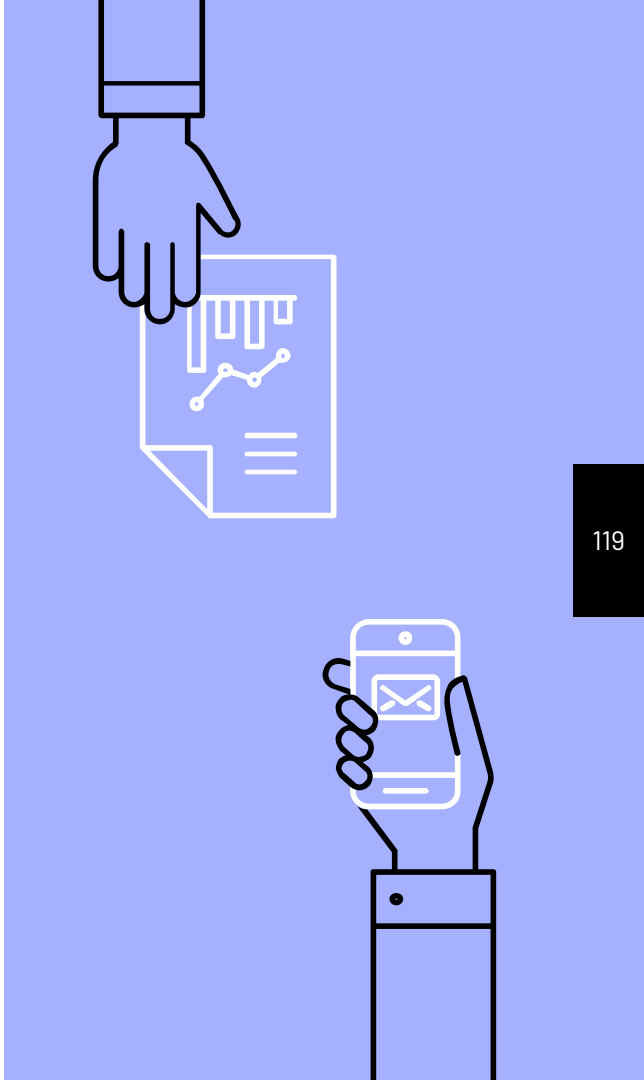
You are the new growth team of a start-up: 18 months of raw data, one question from the board.

- **Diagnose**
  Clean the data, find what really drives (or kills) growth
- **Model**
  Backtest on 2025, project 12 months, 3 scenarios
- **Decide**
  One decision + the A/B test to validate it



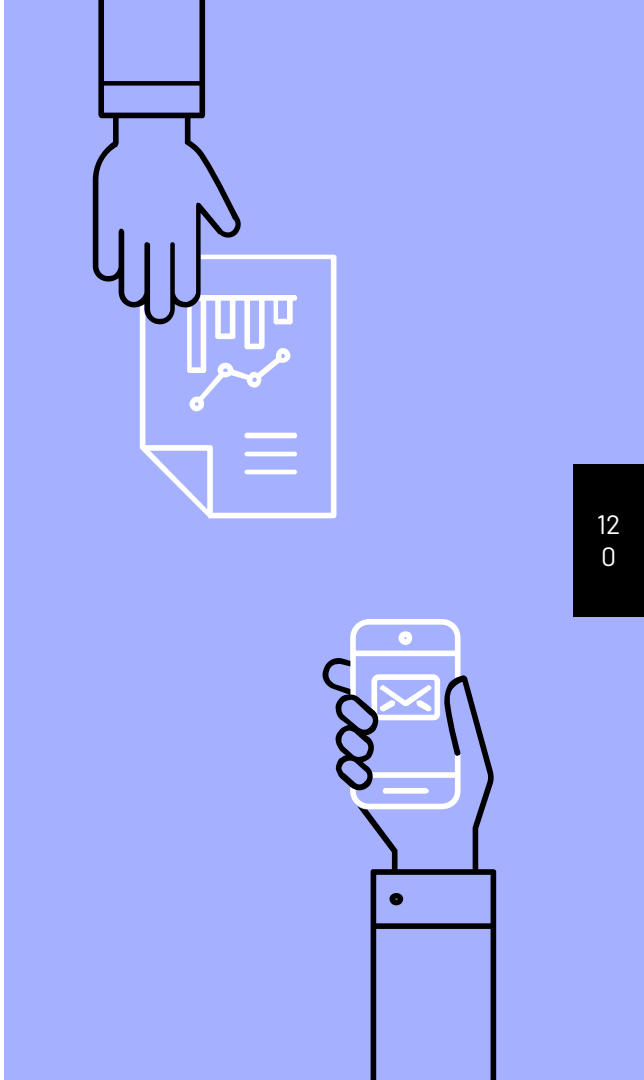
SIX COMPANIES, 23 GROUPS

Company	Growth model	Groups
FitLoop	Freemium fitness app, subscription	1, 7, 13, 19
Kozy	Home-cleaning marketplace, 4 cities	2, 8, 14, 20
Tasko	Product-led B2B SaaS, per seat	3, 9, 15, 21
Verde	DTC e-commerce, plants	4, 10, 16, 22
Splitly	Viral bill-splitting app	5, 11, 17, 23
Echo	Ad-supported podcast app	6, 12, 18



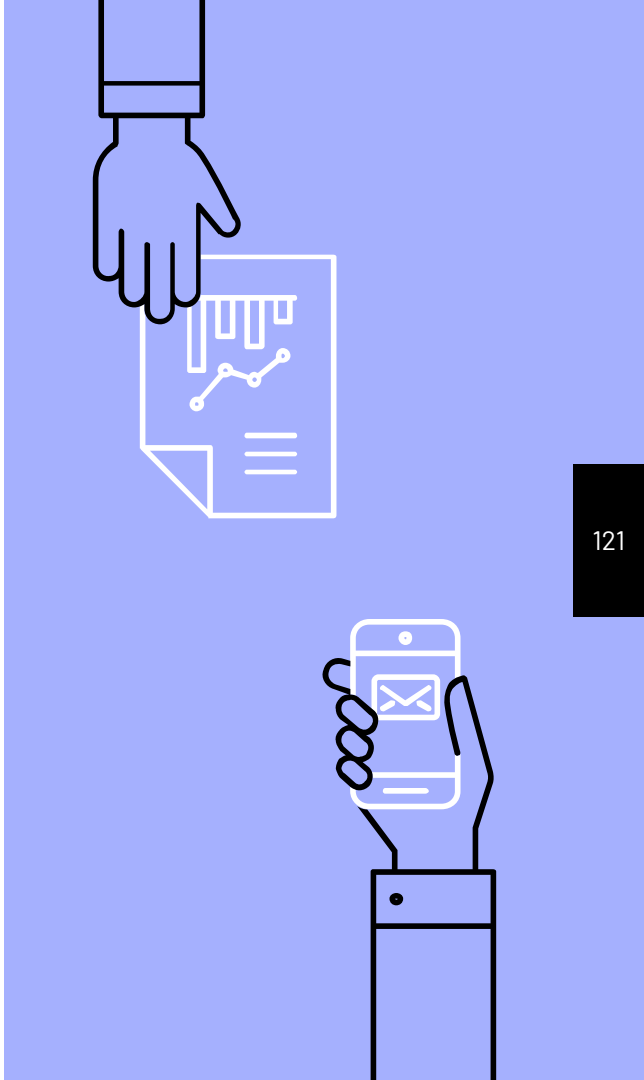
WHAT YOU HAND IN

Deliverable	Format	Must include
Growth model	xlsx or notebook	Backtest + 12 months + 3 scenarios
Report	PDF, 4 pages max	Recommendation, A/B test plan, cleaning log
Slides + defense	10 slides, 8 + 5 min	A cold question, answered live



GRADING (/20)

Criterion	Points
Data cleaning	2
Data diagnosis	5
Growth model and backtest	5
Recommendation and experiment plan	3
Ethics and limits	2
Report	1
Defense and cold question	2



WATCH OUT

- **The data is raw**
  Clean it, and log every rule with the rows it affects
- **Compare like with like**
  Some gaps vanish once you split by channel, city or period
- **Recent cohorts are censored**
  Compare them only on the time they have lived
- **The cold question**
  At the defense, answered live with your own model

